Chapter 3

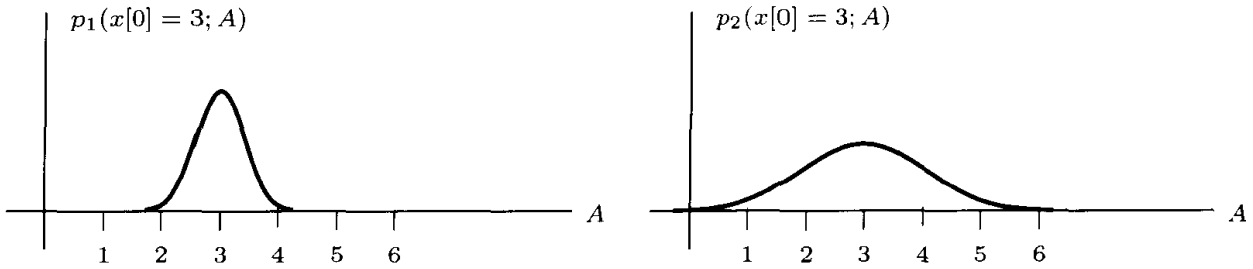
Cramer-Rao Lower Bound

3.1 Introduction

Being able to place a lower bound on the variance of any unbiased estimator proves to be extremely useful in practice. At best, it allows us to assert that an estimator is the MVU estimator. This will be the case if the estimator attains the bound for all values of the unknown parameter. At worst, it provides a benchmark against which we can compare the performance of any unbiased estimator. Furthermore, it alerts us to the physical impossibility of finding an unbiased estimator whose variance is less than the bound. The latter is often useful in signal processing feasibility studies. Although many such variance bounds exist [McAulay and Hofstetter 1971, Kendall and Stuart 1979, Seidman 1970, Ziv and Zakai 1969], the Cramer-Rao lower bound (CRLB) is by far the easiest to determine. Also, the theory allows us to immediately determine if an estimator exists that attains the bound. If no such estimator exists, then all is not lost since estimators can be found that attain the bound in an approximate sense, as described in Chapter 7. For these reasons we restrict our discussion to the CRLB.

3.2 Summary

The CRLB for a scalar parameter is given by (3.6). If the condition (3.7) is satisfied, then the bound will be attained and the estimator that attains it is readily found. An alternative means of determining the CRLB is given by (3.12). For a signal with an unknown parameter in WGN, (3.14) provides a convenient means to evaluate the bound. When a function of a parameter is to be estimated, the CRLB is given by (3.16). Even though an efficient estimator may exist for θ, in general there will not be one for a function of θ (unless the function is linear). For a vector parameter the CRLB is determined using (3.20) and (3.21). As in the scalar parameter case, if condition (3.25) holds, then the bound is attained and the estimator that attains the bound is easily found. For a function of a vector parameter (3.30) provides the bound. A general formula for the Fisher information matrix (used to determine the vector CRLB) for a multivariate Gaussian PDF is given by (3.31). Finally, if the data set comes from a



(a) $\sigma_1 = 1/3$ **(b)** $\sigma_2 = 1$

Figure 3.1 PDF dependence on unknown parameter

WSS Gaussian random process, then an approximate CRLB, that depends on the PSD, is given by (3.34). It is valid asymptotically or as the data record length becomes large.

3.3 Estimator Accuracy Considerations

Before stating the CRLB theorem, it is worthwhile to expose the hidden factors that determine how well we can estimate a parameter. Since all our information is embodied in the observed data and the underlying PDF for that data, it is not surprising that the estimation accuracy depends directly on the PDF. For instance, we should not expect to be able to estimate a parameter with any degree of accuracy if the PDF depends only weakly upon that parameter, or in the extreme case, if the PDF does not depend on it at all. In general, the more the PDF is influenced by the unknown parameter, the better we should be able to estimate it.

Example 3.1 - PDF Dependence on Unknown Parameter

If a single sample is observed as

$$x[0] = A + w[0]$$

where $w[0] \sim \mathcal{N}(0, \sigma^2)$, and it is desired to estimate A, then we expect a better estimate if σ^2 is small. Indeed, a good unbiased estimator is $\hat{A} = x[0]$. The variance is, of course, just σ^2, so that the estimator accuracy improves as σ^2 decreases. An alternative way of viewing this is shown in Figure 3.1, where the PDFs for two different variances are shown. They are

$$p_i(x[0]; A) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left[-\frac{1}{2\sigma_i^2}(x[0] - A)^2\right] \tag{3.1}$$

for $i = 1, 2$. The PDF has been plotted versus the *unknown parameter* A for a *given value of* $x[0]$. If $\sigma_1^2 < \sigma_2^2$, then we should be able to estimate A more accurately based on $p_1(x[0]; A)$. We may interpret this result by referring to Figure 3.1. If $x[0] = 3$ and $\sigma_1 = 1/3$, then as shown in Figure 3.1a, the values of $A > 4$ are highly unlikely. To see

this we determine the probability of observing $x[0]$ in the interval $[x[0]-\delta/2, x[0]+\delta/2] = [3-\delta/2, 3+\delta/2]$ when A takes on a given value or

$$\Pr\left\{3-\frac{\delta}{2} \leq x[0] \leq 3+\frac{\delta}{2}\right\} = \int_{3-\frac{\delta}{2}}^{3+\frac{\delta}{2}} p_i(u;A)\,du$$

which for δ small is $p_i(x[0]=3;A)\delta$. But $p_1(x[0]=3;A=4)\delta = 0.01\delta$, while $p_1(x[0]=3;A=3)\delta = 1.20\delta$. The probability of observing $x[0]$ in a small interval centered about $x[0]=3$ when $A=4$ is small with respect to that when $A=3$. Hence, the values of $A>4$ can be eliminated from consideration. It might be argued that values of A in the interval $3\pm 3\sigma_1 = [2,4]$ are viable candidates. For the PDF in Figure 3.1b there is a much weaker dependence on A. Here our viable candidates are in the much wider interval $3\pm 3\sigma_2 = [0,6]$. ◇

When the PDF is viewed as a function of the unknown parameter (with $\mathbf{x}$ fixed), it is termed the *likelihood function.* Two examples of likelihood functions were shown in Figure 3.1. Intuitively, the "sharpness" of the likelihood function determines how accurately we can estimate the unknown parameter. To quantify this notion observe that the sharpness is effectively measured by the negative of the second derivative of the logarithm of the likelihood function at its peak. This is the *curvature* of the log-likelihood function. In Example 3.1, if we consider the natural logarithm of the PDF

$$\ln p(x[0];A) = -\ln\sqrt{2\pi\sigma^2} - \frac{1}{2\sigma^2}(x[0]-A)^2$$

then the first derivative is

$$\frac{\partial \ln p(x[0];A)}{\partial A} = \frac{1}{\sigma^2}(x[0]-A) \tag{3.2}$$

and the negative of the second derivative becomes

$$-\frac{\partial^2 \ln p(x[0];A)}{\partial A^2} = \frac{1}{\sigma^2}. \tag{3.3}$$

The curvature increases as σ^2 decreases. Since we already know that the estimator $\hat{A}=x[0]$ has variance σ^2, then for this example

$$\mathrm{var}(\hat{A}) = \frac{1}{-\dfrac{\partial^2 \ln p(x[0];A)}{\partial A^2}} \tag{3.4}$$

and the variance decreases as the curvature increases. Although in this example the second derivative does not depend on $x[0]$, in general it will. Thus, a more appropriate measure of curvature is

$$-E\left[\frac{\partial^2 \ln p(x[0];A)}{\partial A^2}\right] \tag{3.5}$$

which measures the *average* curvature of the log-likelihood function. The expectation is taken with respect to $p(x[0];A)$, resulting in a function of A only. The expectation acknowledges the fact that the likelihood function, which depends on $x[0]$, is itself a random variable. The larger the quantity in (3.5), the smaller the variance of the estimator.

3.4 Cramer-Rao Lower Bound

We are now ready to state the CRLB theorem.

Theorem 3.1 (Cramer-Rao Lower Bound - Scalar Parameter) *It is assumed that the PDF $p(\mathbf{x};\theta)$ satisfies the "regularity" condition*

$$E\left[\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right] = 0 \qquad \text{for all } \theta$$

where the expectation is taken with respect to $p(\mathbf{x};\theta)$. Then, the variance of any unbiased estimator $\hat{\theta}$ must satisfy

$$\text{var}(\hat{\theta}) \geq \frac{1}{-E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right]} \tag{3.6}$$

where the derivative is evaluated at the true value of θ and the expectation is taken with respect to $p(\mathbf{x};\theta)$. Furthermore, an unbiased estimator may be found that attains the bound for all θ if and only if

$$\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} = I(\theta)(g(\mathbf{x}) - \theta) \tag{3.7}$$

for some functions g and I. That estimator, which is the MVU estimator, is $\hat{\theta} = g(\mathbf{x})$, and the minimum variance is $1/I(\theta)$.

The expectation in (3.6) is explicitly given by

$$E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right] = \int \frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2} p(\mathbf{x};\theta)\, d\mathbf{x}$$

since the second derivative is a random variable dependent on $\mathbf{x}$. Also, the bound will depend on θ in general, so that it is displayed as in Figure 2.5 (dashed curve). An example of a PDF that does not satisfy the regularity condition is given in Problem 3.1. For a proof of the theorem see Appendix 3A.

Some examples are now given to illustrate the evaluation of the CRLB.

Example 3.2 - CRLB for Example 3.1

For Example 3.1 we see that from (3.3) and (3.6)

$$\text{var}(\hat{A}) \geq \sigma^2 \qquad \text{for all } A.$$

Thus, no unbiased estimator can exist whose variance is lower than σ^2 for even a single value of A. But in fact we know that if $\hat{A} = x[0]$, then $\text{var}(\hat{A}) = \sigma^2$ for all A. Since $x[0]$ is unbiased and attains the CRLB, it must therefore be the MVU estimator. Had we been unable to guess that $x[0]$ would be a good estimator, we could have used (3.7). From (3.2) and (3.7) we make the identification

$$\begin{aligned} \theta &= A \\ I(\theta) &= \frac{1}{\sigma^2} \\ g(x[0]) &= x[0] \end{aligned}$$

so that (3.7) is satisfied. Hence, $\hat{A} = g(x[0]) = x[0]$ is the MVU estimator. Also, note that $\text{var}(\hat{A}) = \sigma^2 = 1/I(\theta)$, so that according to (3.6) we must have

$$I(\theta) = -E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right].$$

We will return to this after the next example. See also Problem 3.2 for a generalization to the non-Gaussian case. ◇

Example 3.3 - DC Level in White Gaussian Noise

Generalizing Example 3.1, consider the multiple observations

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with variance σ^2. To determine the CRLB for A

$$\begin{aligned} p(\mathbf{x};A) &= \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}(x[n]-A)^2\right] \\ &= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)^2\right]. \end{aligned}$$

Taking the first derivative

$$\begin{aligned} \frac{\partial \ln p(\mathbf{x};A)}{\partial A} &= \frac{\partial}{\partial A}\left[-\ln[(2\pi\sigma^2)^{\frac{N}{2}}] - \frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)^2\right] \\ &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1}(x[n]-A) \\ &= \frac{N}{\sigma^2}(\bar{x}-A) \end{aligned} \tag{3.8}$$

where $\bar{x}$ is the sample mean. Differentiating again

$$\frac{\partial^2 \ln p(\mathbf{x}; A)}{\partial A^2} = -\frac{N}{\sigma^2}$$

and noting that the second derivative is a constant, we have from (3.6)

$$\text{var}(\hat{A}) \geq \frac{\sigma^2}{N} \tag{3.9}$$

as the CRLB. Also, by comparing (3.7) and (3.8) we see that the sample mean estimator attains the bound and must therefore be the MVU estimator. Also, once again the minimum variance is given by the reciprocal of the constant N/σ^2 in (3.8). (See also Problems 3.3–3.5 for variations on this example.) ◇

We now prove that when the CRLB is attained

$$\text{var}(\hat{\theta}) = \frac{1}{I(\theta)}$$

where

$$I(\theta) = -E\left[\frac{\partial^2 \ln p(\mathbf{x}; \theta)}{\partial \theta^2}\right].$$

From (3.6) and (3.7)

$$\text{var}(\hat{\theta}) = \frac{1}{-E\left[\frac{\partial^2 \ln p(\mathbf{x}; \theta)}{\partial \theta^2}\right]}$$

and

$$\frac{\partial \ln p(\mathbf{x}; \theta)}{\partial \theta} = I(\theta)(\hat{\theta} - \theta).$$

Differentiating the latter produces

$$\frac{\partial^2 \ln p(\mathbf{x}; \theta)}{\partial \theta^2} = \frac{\partial I(\theta)}{\partial \theta}(\hat{\theta} - \theta) - I(\theta)$$

and taking the negative expected value yields

$$\begin{aligned} -E\left[\frac{\partial^2 \ln p(\mathbf{x}; \theta)}{\partial \theta^2}\right] &= -\frac{\partial I(\theta)}{\partial \theta}[E(\hat{\theta}) - \theta] + I(\theta) \\ &= I(\theta) \end{aligned}$$

and therefore

$$\text{var}(\hat{\theta}) = \frac{1}{I(\theta)}. \tag{3.10}$$

In the next example we will see that the CRLB is not always satisfied.

Example 3.4 - Phase Estimation

Assume that we wish to estimate the phase ϕ of a sinusoid embedded in WGN or

$$x[n] = A\cos(2\pi f_0 n + \phi) + w[n] \qquad n = 0, 1, \ldots, N-1.$$

The amplitude A and frequency f_0 are assumed known (see Example 3.14 for the case when they are unknown). The PDF is

$$p(\mathbf{x};\phi) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left\{-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}[x[n] - A\cos(2\pi f_0 n + \phi)]^2\right\}.$$

Differentiating the log-likelihood function produces

$$\begin{aligned}\frac{\partial \ln p(\mathbf{x};\phi)}{\partial\phi} &= -\frac{1}{\sigma^2}\sum_{n=0}^{N-1}[x[n] - A\cos(2\pi f_0 n + \phi)]A\sin(2\pi f_0 n + \phi) \\ &= -\frac{A}{\sigma^2}\sum_{n=0}^{N-1}[x[n]\sin(2\pi f_0 n + \phi) - \frac{A}{2}\sin(4\pi f_0 n + 2\phi)]\end{aligned}$$

and

$$\frac{\partial^2 \ln p(\mathbf{x};\phi)}{\partial\phi^2} = -\frac{A}{\sigma^2}\sum_{n=0}^{N-1}[x[n]\cos(2\pi f_0 n + \phi) - A\cos(4\pi f_0 n + 2\phi)].$$

Upon taking the negative expected value we have

$$\begin{aligned}-E\left[\frac{\partial^2 \ln p(\mathbf{x};\phi)}{\partial\phi^2}\right] &= \frac{A}{\sigma^2}\sum_{n=0}^{N-1}[A\cos^2(2\pi f_0 n + \phi) - A\cos(4\pi f_0 n + 2\phi)] \\ &= \frac{A^2}{\sigma^2}\sum_{n=0}^{N-1}\left[\frac{1}{2} + \frac{1}{2}\cos(4\pi f_0 n + 2\phi) - \cos(4\pi f_0 n + 2\phi)\right] \\ &\approx \frac{NA^2}{2\sigma^2}\end{aligned}$$

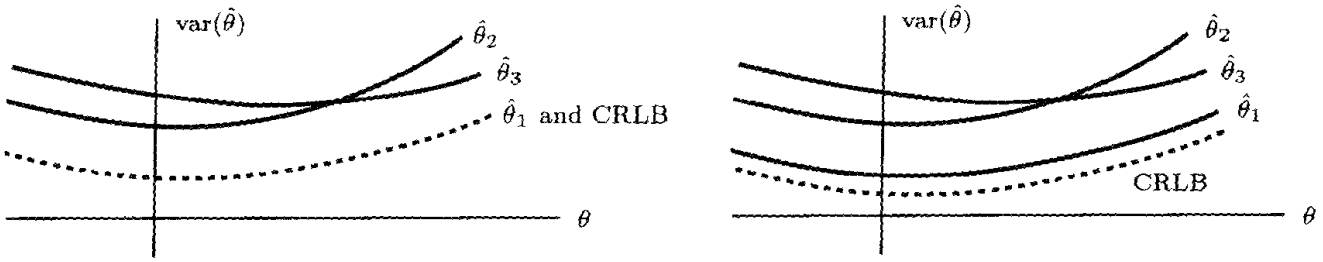
since

$$\frac{1}{N}\sum_{n=0}^{N-1}\cos(4\pi f_0 n + 2\phi) \approx 0$$

for f_0 not near 0 or 1/2 (see Problem 3.7). Therefore,

$$\text{var}(\hat{\phi}) \geq \frac{2\sigma^2}{NA^2}.$$

In this example the condition for the bound to hold is not satisfied. Hence, a phase estimator does not exist which is unbiased and attains the CRLB. It is still possible, however, that an MVU estimator may exist. At this point we do not know how to



(a) $\hat{\theta}_1$ efficient and MVU

(b) $\hat{\theta}_1$ MVU but not efficient

Figure 3.2 Efficiency vs. minimum variance

determine whether an MVU estimator exists, and if it does, how to find it. The theory of sufficient statistics presented in Chapter 5 will allow us to answer these questions. ◇

An estimator which is unbiased and attains the CRLB, as the sample mean estimator in Example 3.3 does, is said to be *efficient* in that it efficiently uses the data. An MVU estimator may or may not be efficient. For instance, in Figure 3.2 the variances of all possible estimators (for purposes of illustration there are three unbiased estimators) are displayed. In Figure 3.2a, $\hat{\theta}_1$ is efficient in that it attains the CRLB. Therefore, it is also the MVU estimator. On the other hand, in Figure 3.2b, $\hat{\theta}_1$ does not attain the CRLB, and hence it is not efficient. However, since its variance is uniformly less than that of all other unbiased estimators, it is the MVU estimator.

The CRLB given by (3.6) may also be expressed in a slightly different form. Although (3.6) is usually more convenient for evaluation, the alternative form is sometimes useful for theoretical work. It follows from the identity (see Appendix 3A)

$$E\left[\left(\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right)^2\right] = -E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right] \tag{3.11}$$

so that

$$\text{var}(\hat{\theta}) \geq \frac{1}{E\left[\left(\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right)^2\right]} \tag{3.12}$$

(see Problem 3.8).

The denominator in (3.6) is referred to as the *Fisher information* $I(\theta)$ for the data $\mathbf{x}$ or

$$I(\theta) = -E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right]. \tag{3.13}$$

As we saw previously, when the CRLB is attained, the variance is the reciprocal of the Fisher information. Intuitively, the more information, the lower the bound. It has the essential properties of an information measure in that it is

1. nonnegative due to (3.11)

2. additive for independent observations.

The latter property leads to the result that the CRLB for N IID observations is $1/N$ times that for one observation. To verify this, note that for independent observations

$$\ln p(\mathbf{x};\theta) = \sum_{n=0}^{N-1} \ln p(x[n];\theta).$$

This results in

$$-E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right] = -\sum_{n=0}^{N-1} E\left[\frac{\partial^2 \ln p(x[n];\theta)}{\partial\theta^2}\right]$$

and finally for identically distributed observations

$$I(\theta) = Ni(\theta)$$

where

$$i(\theta) = -E\left[\frac{\partial^2 \ln p(x[n];\theta)}{\partial\theta^2}\right]$$

is the Fisher information for one sample. For nonindependent samples we might expect that the information will be less than $Ni(\theta)$, as Problem 3.9 illustrates. For completely dependent samples, as for example, $x[0] = x[1] = \cdots = x[N-1]$, we will have $I(\theta) = i(\theta)$ (see also Problem 3.9). Therefore, additional observations carry no information, and the CRLB will not decrease with increasing data record length.

3.5 General CRLB for Signals in White Gaussian Noise

Since it is common to assume white Gaussian noise, it is worthwhile to derive the CRLB for this case. Later, we will extend this to nonwhite Gaussian noise and a vector parameter as given by (3.31). Assume that a deterministic signal with an unknown parameter θ is observed in WGN as

$$x[n] = s[n;\theta] + w[n] \qquad n = 0, 1, \ldots, N-1.$$

The dependence of the signal on θ is explicitly noted. The likelihood function is

$$p(\mathbf{x};\theta) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left\{-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n] - s[n;\theta])^2\right\}.$$

Differentiating once produces

$$\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta} = \frac{1}{\sigma^2}\sum_{n=0}^{N-1}(x[n] - s[n;\theta])\frac{\partial s[n;\theta]}{\partial\theta}$$

and a second differentiation results in

$$\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2} = \frac{1}{\sigma^2}\sum_{n=0}^{N-1}\left\{(x[n]-s[n;\theta])\frac{\partial^2 s[n;\theta]}{\partial\theta^2} - \left(\frac{\partial s[n;\theta]}{\partial\theta}\right)^2\right\}.$$

Taking the expected value yields

$$E\left(\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right) = -\frac{1}{\sigma^2}\sum_{n=0}^{N-1}\left(\frac{\partial s[n;\theta]}{\partial\theta}\right)^2$$

so that finally

$$\text{var}(\hat{\theta}) \geq \frac{\sigma^2}{\sum_{n=0}^{N-1}\left(\frac{\partial s[n;\theta]}{\partial\theta}\right)^2}. \tag{3.14}$$

The form of the bound demonstrates the importance of the signal dependence on θ. Signals that change rapidly as the unknown parameter changes result in accurate estimators. A simple application of (3.14) to Example 3.3, in which $s[n;\theta] = \theta$, produces a CRLB of σ^2/N. The reader should also verify the results of Example 3.4. As a final example we examine the problem of frequency estimation.

Example 3.5 - Sinusoidal Frequency Estimation

We assume that the signal is sinusoidal and is represented as

$$s[n;f_0] = A\cos(2\pi f_0 n + \phi) \qquad 0 < f_0 < \frac{1}{2}$$

where the amplitude and phase are known (see Example 3.14 for the case when they are unknown). From (3.14) the CRLB becomes

$$\text{var}(\hat{f}_0) \geq \frac{\sigma^2}{A^2\sum_{n=0}^{N-1}[2\pi n\sin(2\pi f_0 n + \phi)]^2}. \tag{3.15}$$

The CRLB is plotted in Figure 3.3 versus frequency for an SNR of $A^2/\sigma^2 = 1$, a data record length of $N = 10$, and a phase of $\phi = 0$. It is interesting to note that there appear to be preferred frequencies (see also Example 3.14) for an approximation to (3.15)). Also, as $f_0 \to 0$, the CRLB goes to infinity. This is because for f_0 close to zero a slight change in frequency will not alter the signal significantly. $\diamond$

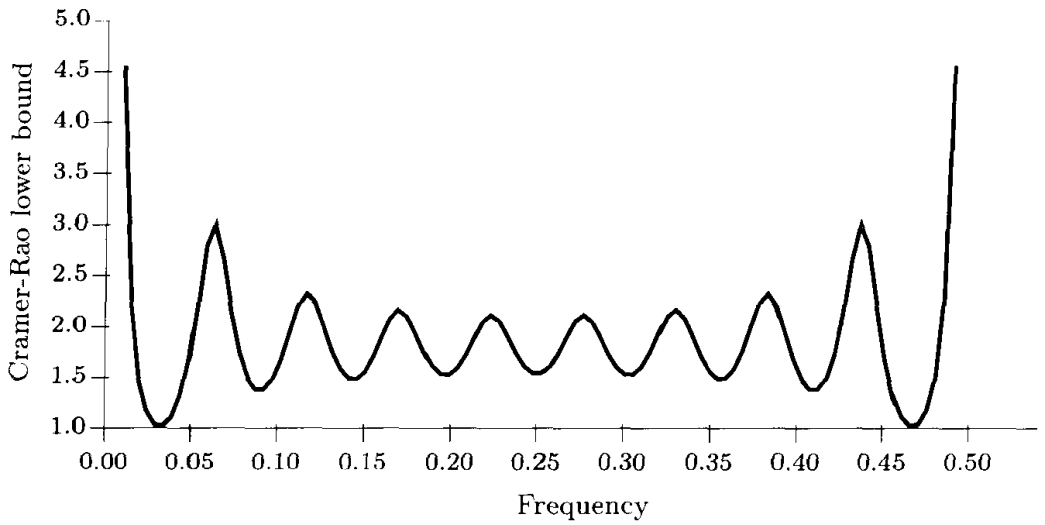


Figure 3.3 Cramer-Rao lower bound for sinusoidal frequency estimation

3.6 Transformation of Parameters

It frequently occurs in practice that the parameter we wish to estimate is a function of some more fundamental parameter. For instance, in Example 3.3 we may not be interested in the sign of A but instead may wish to estimate A^2 or the power of the signal. Knowing the CRLB for A, we can easily obtain it for A^2 or in general for any function of A. As shown in Appendix 3A, if it is desired to estimate $\alpha = g(\theta)$, then the CRLB is

$$\text{var}(\hat{\alpha}) \geq \frac{\left(\frac{\partial g}{\partial \theta}\right)^2}{-E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right]}. \tag{3.16}$$

For the present example this becomes $\alpha = g(A) = A^2$ and

$$\text{var}(\widehat{A^2}) \geq \frac{(2A)^2}{N/\sigma^2} = \frac{4A^2\sigma^2}{N}. \tag{3.17}$$

Note that in using (3.16) the CRLB is expressed in terms of θ.

We saw in Example 3.3 that the sample mean estimator was efficient for A. It might be supposed that $\bar{x}^2$ is efficient for A^2. To quickly dispel this notion we first show that $\bar{x}^2$ is not even an unbiased estimator. Since $\bar{x} \sim \mathcal{N}(A, \sigma^2/N)$

$$\begin{aligned} E(\bar{x}^2) &= E^2(\bar{x}) + \text{var}(\bar{x}) = A^2 + \frac{\sigma^2}{N} \\ &\neq A^2. \end{aligned} \tag{3.18}$$

Hence, we immediately conclude that the *efficiency of an estimator is destroyed by a nonlinear transformation.* That it is maintained for *linear* (actually affine) transformations is easily verified. Assume that an efficient estimator for θ exists and is given

by $\hat{\theta}$. It is desired to estimate $g(\theta) = a\theta + b$. As our estimator of $g(\theta)$, we choose $\widehat{g(\theta)} = g(\hat{\theta}) = a\hat{\theta} + b$. Then,

$$\begin{aligned} E(a\hat{\theta} + b) &= aE(\hat{\theta}) + b = a\theta + b \\ &= g(\theta) \end{aligned}$$

so that $\widehat{g(\theta)}$ is unbiased. The CRLB for $g(\theta)$, is from (3.16),

$$\begin{aligned} \text{var}(\widehat{g(\theta)}) &\geq \frac{\left(\frac{\partial g}{\partial \theta}\right)^2}{I(\theta)} \\ &= \left(\frac{\partial g(\theta)}{\partial \theta}\right)^2 \text{var}(\hat{\theta}) \\ &= a^2 \text{var}(\hat{\theta}). \end{aligned}$$

But $\text{var}(\widehat{g(\theta)}) = \text{var}(a\hat{\theta} + b) = a^2\text{var}(\hat{\theta})$, so that the CRLB is achieved.

Although efficiency is preserved only over linear transformations, it is *approximately* maintained over nonlinear transformations *if the data record is large enough.* This has great practical significance in that we are frequently interested in estimating functions of parameters. To see why this property holds, we return to the previous example of estimating A^2 by $\bar{x}^2$. Although $\bar{x}^2$ is biased, we note from (3.18) that $\bar{x}^2$ is *asymptotically* unbiased or unbiased as $N \to \infty$. Furthermore, since $\bar{x} \sim \mathcal{N}(A, \sigma^2/N)$, we can evaluate the variance

$$\text{var}(\bar{x}^2) = E(\bar{x}^4) - E^2(\bar{x}^2)$$

by using the result that if $\xi \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\begin{aligned} E(\xi^2) &= \mu^2 + \sigma^2 \\ E(\xi^4) &= \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4 \end{aligned}$$

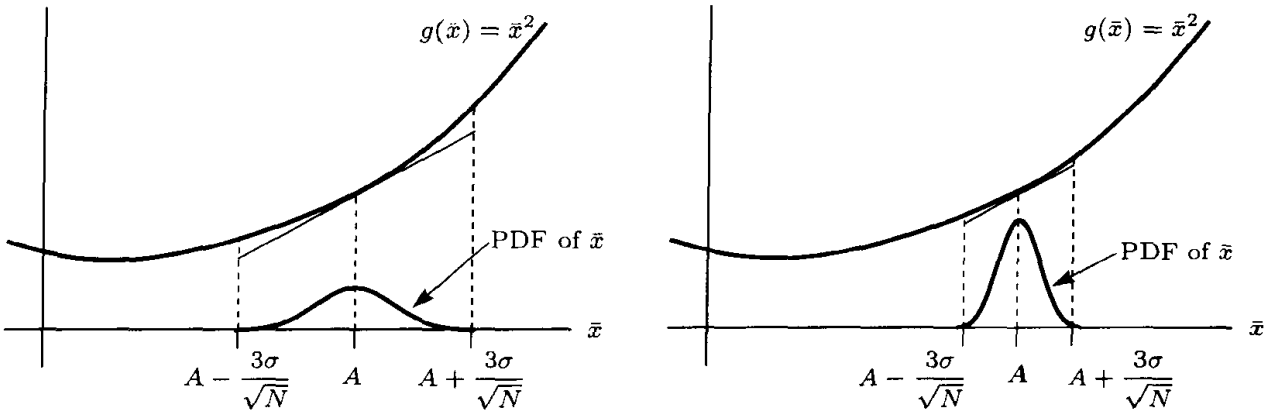
and therefore

$$\begin{aligned} \text{var}(\xi^2) &= E(\xi^4) - E^2(\xi^2) \\ &= 4\mu^2\sigma^2 + 2\sigma^4. \end{aligned}$$

For our problem we have then

$$\text{var}(\bar{x}^2) = \frac{4A^2\sigma^2}{N} + \frac{2\sigma^4}{N^2}. \tag{3.19}$$

Hence, as $N \to \infty$, the variance approaches $4A^2\sigma^2/N$, the last term in (3.19) converging to zero faster than the first. But this is just the CRLB as given by (3.17). Our assertion that $\bar{x}^2$ is an *asymptotically* efficient estimator of A^2 is verified. Intuitively, this situation occurs due to the *statistical linearity* of the transformation, as illustrated in Figure 3.4. As N increases, the PDF of $\bar{x}$ becomes more concentrated about the mean A. Therefore,



(a) Small N **(b)** Large N

Figure 3.4 Statistical linearity of nonlinear transformations

the values of $\bar{x}$ that are observed lie in a small interval about $\bar{x} = A$ (the ± 3 standard deviation interval is displayed). Over this small interval the nonlinear transformation is approximately linear. Therefore, the transformation may be replaced by a linear one since a value of $\bar{x}$ in the *nonlinear region* rarely occurs. In fact, if we linearize g about A, we have the approximation

$$g(\bar{x}) \approx g(A) + \frac{dg(A)}{dA}(\bar{x} - A).$$

It follows that, to within this approximation,

$$E[g(\bar{x})] = g(A) = A^2$$

or the estimator is unbiased (asymptotically). Also,

$$\begin{aligned} \text{var}[g(\bar{x})] &= \left[\frac{dg(A)}{dA}\right]^2 \text{var}(\bar{x}) \\ &= \frac{(2A)^2\sigma^2}{N} \\ &= \frac{4A^2\sigma^2}{N} \end{aligned}$$

so that the estimator achieves the CRLB (asymptotically). Therefore, it is *asymptotically efficient.* This result also yields insight into the form of the CRLB given by (3.16).

3.7 Extension to a Vector Parameter

We now extend the results of the previous sections to the case where we wish to estimate a *vector* parameter $\boldsymbol{\theta} = [\theta_1\ \theta_2 \ldots \theta_p]^T$. We will assume that the estimator $\hat{\boldsymbol{\theta}}$ is unbiased

as defined in Section 2.7. The vector parameter CRLB will allow us to place a bound on the variance of each element. As derived in Appendix 3B, the CRLB is found as the $[i,i]$ element of the inverse of a matrix or

$$\text{var}(\hat{\theta}_i) \geq [\mathbf{I}^{-1}(\boldsymbol{\theta})]_{ii} \tag{3.20}$$

where $\mathbf{I}(\boldsymbol{\theta})$ is the $p \times p$ *Fisher information matrix*. The latter is defined by

$$[\mathbf{I}(\boldsymbol{\theta})]_{ij} = -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\theta_i \partial\theta_j}\right] \tag{3.21}$$

for $i = 1,2,\ldots,p; j = 1,2,\ldots,p$. In evaluating (3.21) the true value of $\boldsymbol{\theta}$ is used. Note that in the scalar case ($p = 1$), $\mathbf{I}(\boldsymbol{\theta}) = I(\theta)$ and we have the scalar CRLB. Some examples follow.

Example 3.6 - DC Level in White Gaussian Noise (Revisited)

We now extend Example 3.3 to the case where in addition to A the noise variance σ^2 is also unknown. The parameter vector is $\boldsymbol{\theta} = [A\ \sigma^2]^T$, and hence $p = 2$. The 2×2 Fisher information matrix is

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A^2}\right] & -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A \partial\sigma^2}\right] \\ -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\sigma^2 \partial A}\right] & -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\sigma^{2\,2}}\right] \end{bmatrix}.$$

It is clear from (3.21) that the matrix is symmetric since the order of partial differentiation may be interchanged and can also be shown to be positive definite (see Problem 3.10). The log-likelihood function is, from Example 3.3,

$$\ln p(\mathbf{x};\boldsymbol{\theta}) = -\frac{N}{2}\ln 2\pi - \frac{N}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)^2.$$

The derivatives are easily found as

$$\begin{aligned}
\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A} &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1}(x[n]-A) \\
\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\sigma^2} &= -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{n=0}^{N-1}(x[n]-A)^2 \\
\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A^2} &= -\frac{N}{\sigma^2} \\
\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A \partial\sigma^2} &= -\frac{1}{\sigma^4}\sum_{n=0}^{N-1}(x[n]-A) \\
\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\sigma^{2\,2}} &= \frac{N}{2\sigma^4} - \frac{1}{\sigma^6}\sum_{n=0}^{N-1}(x[n]-A)^2.
\end{aligned}$$

Upon taking the negative expectations, the Fisher information matrix becomes

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} \frac{N}{\sigma^2} & 0 \\ 0 & \frac{N}{2\sigma^4} \end{bmatrix}.$$

Although not true in general, for this example the Fisher information matrix is diagonal and hence easily inverted to yield

$$\begin{aligned} \text{var}(\hat{A}) &\geq \frac{\sigma^2}{N} \\ \text{var}(\hat{\sigma^2}) &\geq \frac{2\sigma^4}{N}. \end{aligned}$$

Note that the CRLB for $\hat{A}$ is the same as for the case when σ^2 is known due to the diagonal nature of the matrix. Again this is not true in general, as the next example illustrates. ◇

Example 3.7 - Line Fitting

Consider the problem of line fitting or given the observations

$$x[n] = A + Bn + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN, determine the CRLB for the slope B and the intercept A. The parameter vector in this case is $\boldsymbol{\theta} = [A\ B]^T$. We need to first compute the 2×2 Fisher information matrix,

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A^2}\right] & -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A \partial B}\right] \\ -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial B \partial A}\right] & -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial B^2}\right] \end{bmatrix}.$$

The likelihood function is

$$p(\mathbf{x};\boldsymbol{\theta}) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left\{-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n] - A - Bn)^2\right\}$$

from which the derivatives follow as

$$\begin{aligned} \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A} &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1}(x[n] - A - Bn) \\ \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial B} &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1}(x[n] - A - Bn)n \end{aligned}$$

and

$$\begin{aligned}
\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A^2} &= -\frac{N}{\sigma^2} \\
\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A \partial B} &= -\frac{1}{\sigma^2}\sum_{n=0}^{N-1} n \\
\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial B^2} &= -\frac{1}{\sigma^2}\sum_{n=0}^{N-1} n^2.
\end{aligned}$$

Since the second-order derivatives do not depend on $\mathbf{x}$, we have immediately that

$$\begin{aligned}
\mathbf{I}(\boldsymbol{\theta}) &= \frac{1}{\sigma^2}\begin{bmatrix} N & \sum_{n=0}^{N-1} n \\ \sum_{n=0}^{N-1} n & \sum_{n=0}^{N-1} n^2 \end{bmatrix} \\
&= \frac{1}{\sigma^2}\begin{bmatrix} N & \frac{N(N-1)}{2} \\ \frac{N(N-1)}{2} & \frac{N(N-1)(2N-1)}{6} \end{bmatrix}
\end{aligned}$$

where we have used the identities

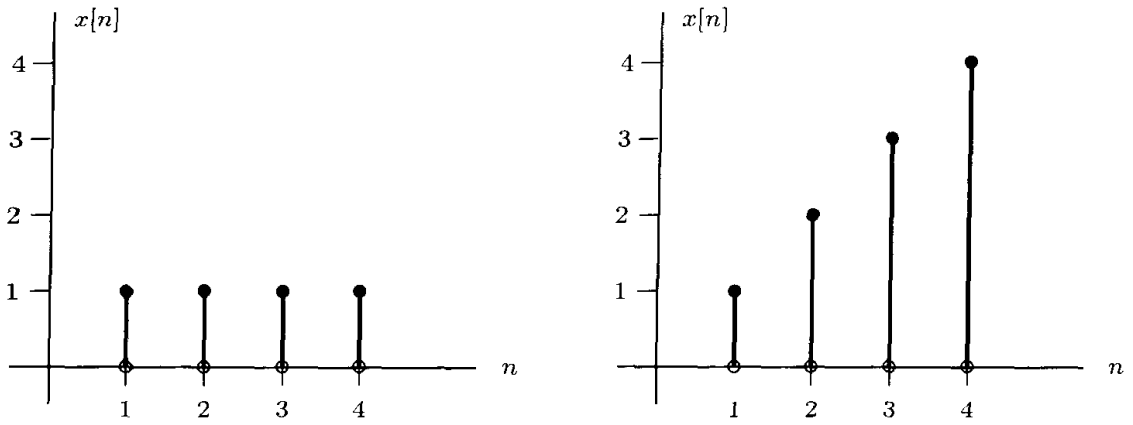
$$\begin{aligned}
\sum_{n=0}^{N-1} n &= \frac{N(N-1)}{2} \\
\sum_{n=0}^{N-1} n^2 &= \frac{N(N-1)(2N-1)}{6}.
\end{aligned} \tag{3.22}$$

Inverting the matrix yields

$$\mathbf{I}^{-1}(\boldsymbol{\theta}) = \sigma^2 \begin{bmatrix} \frac{2(2N-1)}{N(N+1)} & -\frac{6}{N(N+1)} \\ -\frac{6}{N(N+1)} & \frac{12}{N(N^2-1)} \end{bmatrix}.$$

It follows from (3.20) that the CRLB is

$$\begin{aligned}
\mathrm{var}(\hat{A}) &\geq \frac{2(2N-1)\sigma^2}{N(N+1)} \\
\mathrm{var}(\hat{B}) &\geq \frac{12\sigma^2}{N(N^2-1)}.
\end{aligned}$$



(a) $A = 0$, $B = 0$ to $A = 1$, $B = 0$ **(b)** $A = 0$, $B = 0$ to $A = 0$, $B = 1$

Figure 3.5 Sensitivity of observations to parameter changes—no noise

Some interesting observations follow from examination of the CRLB. Note first that the CRLB for $\hat{A}$ has increased over that obtained when B is known, for in the latter case we have

$$\text{var}(\hat{A}) \geq -\frac{1}{E\left[\frac{\partial^2 \ln p(\mathbf{x}; A)}{\partial A^2}\right]} = \frac{\sigma^2}{N}$$

and for $N \geq 2$, $2(2N-1)/(N+1) > 1$. This is a quite general result that asserts that *the CRLB always increases as we estimate more parameters* (see Problems 3.11 and 3.12). A second point is that

$$\frac{\text{CRLB}(\hat{A})}{\text{CRLB}(\hat{B})} = \frac{(2N-1)(N-1)}{6} > 1$$

for $N \geq 3$. Hence, B is easier to estimate, its CRLB decreasing as $1/N^3$ as opposed to the $1/N$ dependence for the CRLB of A. These differing dependences indicate that $x[n]$ is *more sensitive* to changes in B than to changes in A. A simple calculation reveals

$$\begin{aligned} \Delta x[n] &\approx \frac{\partial x[n]}{\partial A}\Delta A = \Delta A \\ \Delta x[n] &\approx \frac{\partial x[n]}{\partial B}\Delta B = n\Delta B. \end{aligned}$$

Changes in B are magnified by n, as illustrated in Figure 3.5. This effect is reminiscent of (3.14), and indeed a similar type of relationship is obtained in the vector parameter case (see (3.33)). See Problem 3.13 for a generalization of this example. ◇

As an alternative means of computing the CRLB we can use the identity

$$E\left[\frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta_i}\frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta_j}\right] = -E\left[\frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta_i \partial \theta_j}\right] \tag{3.23}$$

as shown in Appendix 3B. The form given on the right-hand side is usually easier to evaluate, however.

We now formally state the CRLB theorem for a vector parameter. Included in the theorem are conditions for equality. The bound is stated in terms of the covariance matrix of $\hat{\boldsymbol{\theta}}$, denoted by $\mathbf{C}_{\hat{\theta}}$, from which (3.20) follows.

Theorem 3.2 (Cramer-Rao Lower Bound - Vector Parameter) *It is assumed that the PDF $p(\mathbf{x};\boldsymbol{\theta})$ satisfies the "regularity" conditions*

$$E\left[\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right] = \mathbf{0} \qquad \text{for all } \boldsymbol{\theta}$$

where the expectation is taken with respect to $p(\mathbf{x};\boldsymbol{\theta})$. Then, the covariance matrix of any unbiased estimator $\hat{\boldsymbol{\theta}}$ satisfies

$$\mathbf{C}_{\hat{\theta}} - \mathbf{I}^{-1}(\boldsymbol{\theta}) \geq \mathbf{0} \tag{3.24}$$

where $\geq \mathbf{0}$ is interpreted as meaning that the matrix is positive semidefinite. The Fisher information matrix $\mathbf{I}(\boldsymbol{\theta})$ is given as

$$[\mathbf{I}(\boldsymbol{\theta})]_{ij} = -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j}\right]$$

where the derivatives are evaluated at the true value of $\boldsymbol{\theta}$ and the expectation is taken with respect to $p(\mathbf{x};\boldsymbol{\theta})$. Furthermore, an unbiased estimator may be found that attains the bound in that $\mathbf{C}_{\hat{\theta}} = \mathbf{I}^{-1}(\boldsymbol{\theta})$ if and only if

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{I}(\boldsymbol{\theta})(\mathbf{g}(\mathbf{x}) - \boldsymbol{\theta}) \tag{3.25}$$

for some p-dimensional function $\mathbf{g}$ and some $p \times p$ matrix $\mathbf{I}$. That estimator, which is the MVU estimator, is $\hat{\boldsymbol{\theta}} = \mathbf{g}(\mathbf{x})$, and its covariance matrix is $\mathbf{I}^{-1}(\boldsymbol{\theta})$.

The proof is given in Appendix 3B. That (3.20) follows from (3.24) is shown by noting that for a positive semidefinite matrix the diagonal elements are nonnegative. Hence,

$$\left[\mathbf{C}_{\hat{\theta}} - \mathbf{I}^{-1}(\boldsymbol{\theta})\right]_{ii} \geq 0$$

and therefore

$$\text{var}(\hat{\theta}_i) = [\mathbf{C}_{\hat{\theta}}]_{ii} \geq \left[\mathbf{I}^{-1}(\boldsymbol{\theta})\right]_{ii}. \tag{3.26}$$

Additionally, when equality holds or $\mathbf{C}_{\hat{\theta}} = \mathbf{I}^{-1}(\boldsymbol{\theta})$, then (3.26) holds with equality also. The conditions for the CRLB to be attained are of particular interest since then $\hat{\boldsymbol{\theta}} = \mathbf{g}(\mathbf{x})$ is efficient and hence is the MVU estimator. An example of equality occurs in Example 3.7. There we found that

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \begin{bmatrix} \dfrac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A} \\ \dfrac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial B} \end{bmatrix} \tag{3.27}$$

$$= \begin{bmatrix} \frac{1}{\sigma^2}\sum_{n=0}^{N-1}(x[n]-A-Bn) \\ \frac{1}{\sigma^2}\sum_{n=0}^{N-1}(x[n]-A-Bn)n \end{bmatrix}. \tag{3.28}$$

Although not obvious, this may be rewritten as

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \begin{bmatrix} \frac{N}{\sigma^2} & \frac{N(N-1)}{2\sigma^2} \\ \frac{N(N-1)}{2\sigma^2} & \frac{N(N-1)(2N-1)}{6\sigma^2} \end{bmatrix} \begin{bmatrix} \hat{A}-A \\ \hat{B}-B \end{bmatrix} \tag{3.29}$$

where

$$\begin{aligned} \hat{A} &= \frac{2(2N-1)}{N(N+1)}\sum_{n=0}^{N-1}x[n] - \frac{6}{N(N+1)}\sum_{n=0}^{N-1}nx[n] \\ \hat{B} &= -\frac{6}{N(N+1)}\sum_{n=0}^{N-1}x[n] + \frac{12}{N(N^2-1)}\sum_{n=0}^{N-1}nx[n]. \end{aligned}$$

Hence, the conditions for equality are satisfied and $[\hat{A}\ \hat{B}]^T$ is an efficient and therefore MVU estimator. Furthermore, the matrix in (3.29) is the inverse of the covariance matrix.

If the equality conditions hold, the reader may ask whether we can be assured that $\hat{\boldsymbol{\theta}}$ is unbiased. Because the regularity conditions

$$E\left[\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right] = \mathbf{0}$$

are always assumed to hold, we can apply them to (3.25). This then yields $E[\mathbf{g}(\mathbf{x})] = E(\hat{\boldsymbol{\theta}}) = \boldsymbol{\theta}$.

In finding MVU estimators for a vector parameter the CRLB theorem provides a powerful tool. In particular, it allows us to find the MVU estimator for an important class of data models. This class is the *linear model* and is described in detail in Chapter 4. The line fitting example just discussed is a special case. Suffice it to say that if we can model our data in the linear model form, then the MVU estimator and its performance are easily found.

3.8 Vector Parameter CRLB for Transformations

The discussion in Section 3.6 extends readily to the vector case. Assume that it is desired to estimate $\boldsymbol{\alpha} = \mathbf{g}(\boldsymbol{\theta})$ for $\mathbf{g}$, an r-dimensional function. Then, as shown in Appendix 3B

$$\mathbf{C}_{\hat{\alpha}} - \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\mathbf{I}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{g}(\boldsymbol{\theta})^T}{\partial \boldsymbol{\theta}} \geq \mathbf{0} \tag{3.30}$$

where, as before, $\geq \mathbf{0}$ is to be interpreted as positive semidefinite. In (3.30), $\partial \mathbf{g}(\boldsymbol{\theta})/\partial \boldsymbol{\theta}$ is the $r \times p$ Jacobian matrix defined as

$$\frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \begin{bmatrix} \frac{\partial g_1(\boldsymbol{\theta})}{\partial \theta_1} & \frac{\partial g_1(\boldsymbol{\theta})}{\partial \theta_2} & \cdots & \frac{\partial g_1(\boldsymbol{\theta})}{\partial \theta_p} \\ \frac{\partial g_2(\boldsymbol{\theta})}{\partial \theta_1} & \frac{\partial g_2(\boldsymbol{\theta})}{\partial \theta_2} & \cdots & \frac{\partial g_2(\boldsymbol{\theta})}{\partial \theta_p} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_r(\boldsymbol{\theta})}{\partial \theta_1} & \frac{\partial g_r(\boldsymbol{\theta})}{\partial \theta_2} & \cdots & \frac{\partial g_r(\boldsymbol{\theta})}{\partial \theta_p} \end{bmatrix}.$$

Example 3.8 - CRLB for Signal-to-Noise Ratio

Consider a DC level in WGN with A and σ^2 unknown. We wish to estimate

$$\alpha = \frac{A^2}{\sigma^2}$$

which can be considered to be the SNR for a single sample. Here $\boldsymbol{\theta} = [A\, \sigma^2]^T$ and $g(\boldsymbol{\theta}) = \theta_1^2/\theta_2 = A^2/\sigma^2$. Then, as shown in Example 3.6,

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} \frac{N}{\sigma^2} & 0 \\ 0 & \frac{N}{2\sigma^4} \end{bmatrix}.$$

The Jacobian is

$$\begin{aligned} \frac{\partial g(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} &= \begin{bmatrix} \frac{\partial g(\boldsymbol{\theta})}{\partial \theta_1} & \frac{\partial g(\boldsymbol{\theta})}{\partial \theta_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial g(\boldsymbol{\theta})}{\partial A} & \frac{\partial g(\boldsymbol{\theta})}{\partial \sigma^2} \end{bmatrix} \\ &= \begin{bmatrix} \frac{2A}{\sigma^2} & -\frac{A^2}{\sigma^4} \end{bmatrix} \end{aligned}$$

so that

$$\begin{aligned} \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{g}(\boldsymbol{\theta})^T}{\partial \boldsymbol{\theta}} &= \begin{bmatrix} \frac{2A}{\sigma^2} & -\frac{A^2}{\sigma^4} \end{bmatrix} \begin{bmatrix} \frac{\sigma^2}{N} & 0 \\ 0 & \frac{2\sigma^4}{N} \end{bmatrix} \begin{bmatrix} \frac{2A}{\sigma^2} \\ -\frac{A^2}{\sigma^4} \end{bmatrix} \\ &= \frac{4A^2}{N\sigma^2} + \frac{2A^4}{N\sigma^4} \\ &= \frac{4\alpha + 2\alpha^2}{N}. \end{aligned}$$

Finally, since α is a scalar

$$\mathrm{var}(\hat{\alpha}) \geq \frac{4\alpha + 2\alpha^2}{N}.$$

$\diamond$

As discussed in Section 3.6, efficiency is maintained over linear transformations

$$\boldsymbol{\alpha} = \mathbf{g}(\boldsymbol{\theta}) = \mathbf{A}\boldsymbol{\theta} + \mathbf{b}$$

where $\mathbf{A}$ is an $r \times p$ matrix and $\mathbf{b}$ is an $r \times 1$ vector. If $\hat{\boldsymbol{\alpha}} = \mathbf{A}\hat{\boldsymbol{\theta}} + \mathbf{b}$, and $\hat{\boldsymbol{\theta}}$ is efficient or $\mathbf{C}_{\hat{\theta}} = \mathbf{I}^{-1}(\boldsymbol{\theta})$, then

$$E(\hat{\boldsymbol{\alpha}}) = \mathbf{A}\boldsymbol{\theta} + \mathbf{b} = \boldsymbol{\alpha}$$

so that $\hat{\boldsymbol{\alpha}}$ is unbiased and

$$\begin{aligned} \mathbf{C}_{\hat{\alpha}} &= \mathbf{A}\mathbf{C}_{\hat{\theta}}\mathbf{A}^T = \mathbf{A}\mathbf{I}^{-1}(\boldsymbol{\theta})\mathbf{A}^T \\ &= \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{g}(\boldsymbol{\theta})^T}{\partial \boldsymbol{\theta}}, \end{aligned}$$

the latter being the CRLB. For nonlinear transformations efficiency is maintained only as $N \to \infty$. (This assumes that the PDF of $\hat{\boldsymbol{\theta}}$ becomes concentrated about the true value of $\boldsymbol{\theta}$ as $N \to \infty$ or that $\hat{\boldsymbol{\theta}}$ is consistent.) Again this is due to the statistical linearity of $\mathbf{g}(\boldsymbol{\theta})$ about the true value of $\boldsymbol{\theta}$.

3.9 CRLB for the General Gaussian Case

It is quite convenient at times to have a general expression for the CRLB. In the case of Gaussian observations we can derive the CRLB that generalizes (3.14). Assume that

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}(\boldsymbol{\theta}), \mathbf{C}(\boldsymbol{\theta}))$$

so that both the mean and covariance may depend on $\boldsymbol{\theta}$. Then, as shown in Appendix 3C, the Fisher information matrix is given by

$$\begin{aligned} [\mathbf{I}(\boldsymbol{\theta})]_{ij} &= \left[\frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_i}\right]^T \mathbf{C}^{-1}(\boldsymbol{\theta}) \left[\frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_j}\right] \\ &+ \frac{1}{2}\mathrm{tr}\left[\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_i}\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_j}\right] \end{aligned} \tag{3.31}$$

where

$$\frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_i} = \begin{bmatrix} \frac{\partial [\boldsymbol{\mu}(\boldsymbol{\theta})]_1}{\partial \theta_i} \\ \frac{\partial [\boldsymbol{\mu}(\boldsymbol{\theta})]_2}{\partial \theta_i} \\ \vdots \\ \frac{\partial [\boldsymbol{\mu}(\boldsymbol{\theta})]_N}{\partial \theta_i} \end{bmatrix}$$

and

$$\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_i} = \begin{bmatrix} \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{11}}{\partial \theta_i} & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{12}}{\partial \theta_i} & \cdots & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{1N}}{\partial \theta_i} \\ \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{21}}{\partial \theta_i} & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{22}}{\partial \theta_i} & \cdots & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{2N}}{\partial \theta_i} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{N1}}{\partial \theta_i} & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{N2}}{\partial \theta_i} & \cdots & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{NN}}{\partial \theta_i} \end{bmatrix}.$$

For the scalar parameter case in which

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}(\theta), \mathbf{C}(\theta))$$

this reduces to

$$\begin{aligned} I(\theta) &= \left[\frac{\partial \boldsymbol{\mu}(\theta)}{\partial \theta}\right]^T \mathbf{C}^{-1}(\theta) \left[\frac{\partial \boldsymbol{\mu}(\theta)}{\partial \theta}\right] \\ &\quad + \frac{1}{2}\mathrm{tr}\left[\left(\mathbf{C}^{-1}(\theta)\frac{\partial \mathbf{C}(\theta)}{\partial \theta}\right)^2\right] \end{aligned} \tag{3.32}$$

which generalizes (3.14). We now illustrate the computation with some examples.

Example 3.9 - Parameters of a Signal in White Gaussian Noise

Assume that we wish to estimate a scalar signal parameter θ for the data set

$$x[n] = s[n;\theta] + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN. The covariance matrix is $\mathbf{C} = \sigma^2\mathbf{I}$ and does not depend on θ. The second term in (3.32) is therefore zero. The first term yields

$$\begin{aligned} I(\theta) &= \frac{1}{\sigma^2}\left[\frac{\partial \boldsymbol{\mu}(\theta)}{\partial \theta}\right]^T \left[\frac{\partial \boldsymbol{\mu}(\theta)}{\partial \theta}\right] \\ &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1}\left(\frac{\partial [\boldsymbol{\mu}(\theta)]_n}{\partial \theta}\right)^2 \\ &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1}\left(\frac{\partial s[n;\theta]}{\partial \theta}\right)^2 \end{aligned}$$

which agrees with (3.14). ◇

Generalizing to a vector signal parameter estimated in the presence of WGN, we have from (3.31)

$$[\mathbf{I}(\boldsymbol{\theta})]_{ij} = \left[\frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_i}\right]^T \frac{1}{\sigma^2}\mathbf{I}\left[\frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_j}\right]$$

which yields

$$[\mathbf{I}(\boldsymbol{\theta})]_{ij} = \frac{1}{\sigma^2}\sum_{n=0}^{N-1}\frac{\partial s[n;\boldsymbol{\theta}]}{\partial \theta_i}\frac{\partial s[n;\boldsymbol{\theta}]}{\partial \theta_j} \tag{3.33}$$

as the elements of the Fisher information matrix.

Example 3.10 - Parameter of Noise

Assume that we observe

$$x[n] = w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with unknown variance $\theta = \sigma^2$. Then, according to (3.32), since $\mathbf{C}(\sigma^2) = \sigma^2\mathbf{I}$, we have

$$\begin{aligned} I(\sigma^2) &= \frac{1}{2}\mathrm{tr}\left[\left(\mathbf{C}^{-1}(\sigma^2)\frac{\partial \mathbf{C}(\sigma^2)}{\partial \sigma^2}\right)^2\right] \\ &= \frac{1}{2}\mathrm{tr}\left[\left(\left(\frac{1}{\sigma^2}\right)(\mathbf{I})\right)^2\right] \\ &= \frac{1}{2}\mathrm{tr}\left[\frac{1}{\sigma^4}\mathbf{I}\right] \\ &= \frac{N}{2\sigma^4} \end{aligned}$$

which agrees with the results in Example 3.6. A slightly more complicated example follows. ◇

Example 3.11 - Random DC Level in WGN

Consider the data

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN and A, the DC level, is a Gaussian *random variable* with zero mean and variance σ_A^2. Also, A is independent of $w[n]$. The power of the signal or variance σ_A^2 is the unknown parameter. Then, $\mathbf{x} = [x[0]\, x[1] \ldots x[N-1]]^T$ is Gaussian with zero mean and an $N \times N$ covariance matrix whose $[i, j]$ element is

$$\begin{aligned} \left[\mathbf{C}(\sigma_A^2)\right]_{ij} &= E\left[x[i-1]x[j-1]\right] \\ &= E\left[(A + w[i-1])(A + w[j-1])\right] \\ &= \sigma_A^2 + \sigma^2\delta_{ij}. \end{aligned}$$

Therefore,

$$\mathbf{C}(\sigma_A^2) = \sigma_A^2\mathbf{1}\mathbf{1}^T + \sigma^2\mathbf{I}$$

where $\mathbf{1} = [1\,1\ldots1]^T$. Using Woodbury's identity (see Appendix 1), we have

$$\mathbf{C}^{-1}(\sigma_A^2) = \frac{1}{\sigma^2}\left(\mathbf{I} - \frac{\sigma_A^2}{\sigma^2 + N\sigma_A^2}\mathbf{1}\mathbf{1}^T\right).$$

Also, since

$$\frac{\partial \mathbf{C}(\sigma_A^2)}{\partial \sigma_A^2} = \mathbf{1}\mathbf{1}^T$$

we have that

$$\mathbf{C}^{-1}(\sigma_A^2)\frac{\partial \mathbf{C}(\sigma_A^2)}{\partial \sigma_A^2} = \frac{1}{\sigma^2 + N\sigma_A^2}\mathbf{1}\mathbf{1}^T.$$

Substituting this in (3.32) produces

$$\begin{aligned} I(\sigma_A^2) &= \frac{1}{2}\mathrm{tr}\left[\left(\frac{1}{\sigma^2 + N\sigma_A^2}\right)^2 \mathbf{1}\mathbf{1}^T\mathbf{1}\mathbf{1}^T\right] \\ &= \frac{N}{2}\left(\frac{1}{\sigma^2 + N\sigma_A^2}\right)^2 \mathrm{tr}(\mathbf{1}\mathbf{1}^T) \\ &= \frac{1}{2}\left(\frac{N}{\sigma^2 + N\sigma_A^2}\right)^2 \end{aligned}$$

so that the CRLB is

$$\mathrm{var}(\sigma_A^2) \geq 2\left(\sigma_A^2 + \frac{\sigma^2}{N}\right)^2.$$

Note that even as $N \to \infty$, the CRLB does not decrease below $2\sigma_A^4$. This is because each additional data sample yields the same value of A (see Problem 3.14). $\diamond$

3.10 Asymptotic CRLB for WSS Gaussian Random Processes

At times it is difficult to analytically compute the CRLB using (3.31) due to the need to invert the covariance matrix. Of course, we can always resort to a computer evaluation. An alternative form which can be applied to Gaussian processes that are WSS is very useful. It is easily computed and provides much insight due to its simplified form. The principal drawback is that strictly speaking it is valid only as $N \to \infty$ or *asymptotically*. In practice it provides a good approximation to the true CRLB if the data record length N is much greater than the *correlation time* of the process. The correlation time is defined as the maximum lag k of the ACF $r_{xx}[k] = E[x[n]x[n+k]]$ for which the ACF is essentially nonzero. Hence, for processes with broad PSDs the approximation will be good for moderate length data records, while for narrowband processes longer length data records are required.

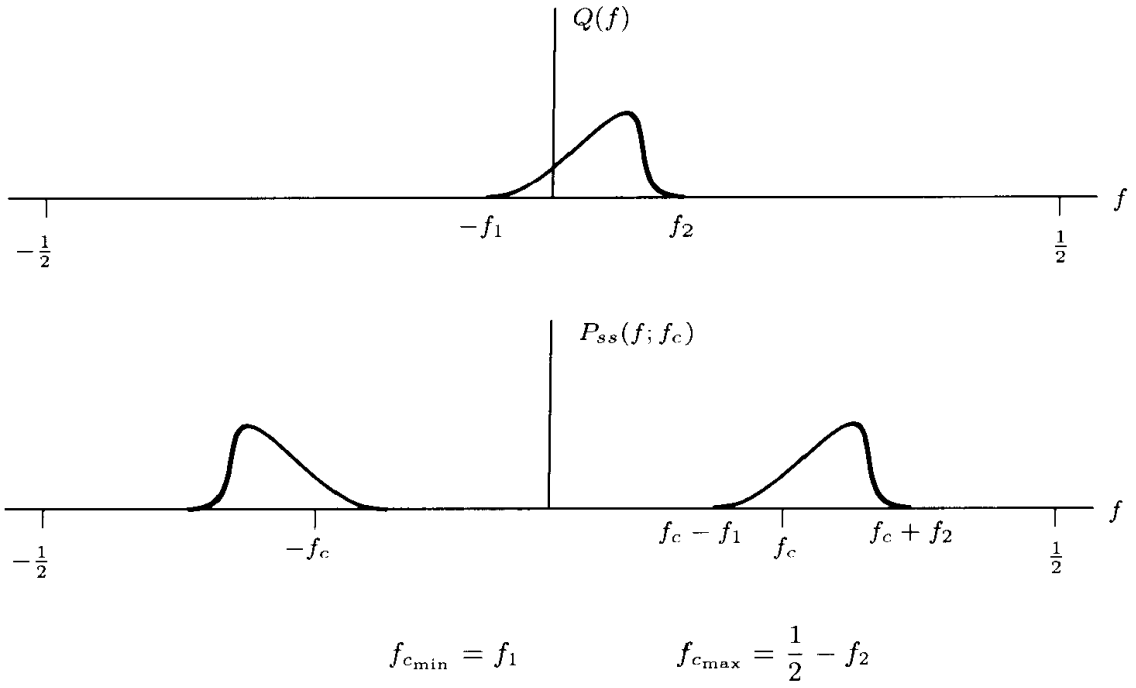


Figure 3.6 Signal PSD for center frequency estimation

As shown in Appendix 3D, the elements of the Fisher information are approximately (as $N \to \infty$)

$$[\mathbf{I}(\boldsymbol{\theta})]_{ij} = \frac{N}{2} \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{\partial \ln P_{xx}(f;\boldsymbol{\theta})}{\partial \theta_i} \frac{\partial \ln P_{xx}(f;\boldsymbol{\theta})}{\partial \theta_j} df \tag{3.34}$$

where $P_{xx}(f;\boldsymbol{\theta})$ is the PSD of the process with the explicit dependence on $\boldsymbol{\theta}$ shown. *It is assumed that the mean of $x[n]$ is zero.* This form, which is somewhat reminiscent of (3.33), allows us to examine the accuracy with which PSD, or equivalently, covariance parameters may be estimated.

Example 3.12 - Center Frequency of Process

A typical problem is to estimate the center frequency f_c of a PSD which otherwise is known. Given

$$P_{xx}(f;f_c) = Q(f - f_c) + Q(-f - f_c) + \sigma^2$$

we wish to determine the CRLB for f_c assuming that $Q(f)$ and σ^2 are known. We may view the process as consisting of a random signal embedded in WGN. The center frequency of the signal PSD is to be estimated. The real function $Q(f)$ and the signal PSD $P_{ss}(f;f_c)$ are shown in Figure 3.6. Note that the possible center frequencies are constrained to be in the interval $[f_1, 1/2 - f_2]$. For these center frequencies the signal PSD for $f \geq 0$ will be contained in the [0,1/2] interval. Then, since $\boldsymbol{\theta} = f_c$ is a scalar,

we have from (3.34)

$$\text{var}(\hat{f}_c) \geq \frac{1}{\frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{\partial \ln P_{xx}(f;f_c)}{\partial f_c}\right)^2 df}.$$

But

$$\begin{aligned}\frac{\partial \ln P_{xx}(f;f_c)}{\partial f_c} &= \frac{\partial \ln [Q(f-f_c)+Q(-f-f_c)+\sigma^2]}{\partial f_c} \\ &= \frac{\frac{\partial Q(f-f_c)}{\partial f_c}+\frac{\partial Q(-f-f_c)}{\partial f_c}}{Q(f-f_c)+Q(-f-f_c)+\sigma^2}.\end{aligned}$$

This is an odd function of f, so that

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{\partial \ln P_{xx}(f;f_c)}{\partial f_c}\right)^2 df = 2\int_0^{\frac{1}{2}} \left(\frac{\partial \ln P_{xx}(f;f_c)}{\partial f_c}\right)^2 df.$$

Also, for $f \geq 0$ we have that $Q(-f-f_c)=0$, and thus its derivative is zero due to the assumption illustrated in Figure 3.6. It follows that

$$\begin{aligned}\text{var}(\hat{f}_c) &\geq \frac{1}{N\int_0^{\frac{1}{2}} \left(\frac{\frac{\partial Q(f-f_c)}{\partial f_c}}{Q(f-f_c)+\sigma^2}\right)^2 df} \\ &= \frac{1}{N\int_0^{\frac{1}{2}} \left(\frac{\frac{\partial Q(f-f_c)}{\partial (f-f_c)}(-1)}{Q(f-f_c)+\sigma^2}\right)^2 df} \\ &= \frac{1}{N\int_{-f_c}^{\frac{1}{2}-f_c} \left(\frac{\frac{\partial Q(f')}{\partial f'}}{Q(f')+\sigma^2}\right)^2 df'}\end{aligned}$$

where we have let $f' = f - f_c$. But $1/2 - f_c \geq 1/2 - f_{c_{\max}} = f_2$ and $-f_c \leq -f_{c_{\min}} = -f_1$, so that we may change the limits of integration to the interval $[-1/2, 1/2]$. Thus,

$$\begin{aligned}\text{var}(\hat{f}_c) &\geq \frac{1}{N\int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{\frac{\partial Q(f)}{\partial f}}{Q(f)+\sigma^2}\right)^2 df} \\ &= \frac{1}{N\int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{\partial \ln(Q(f)+\sigma^2)}{\partial f}\right)^2 df}.\end{aligned}$$

As an example, consider

$$Q(f) = \exp\left[-\frac{1}{2}\left(\frac{f}{\sigma_f}\right)^2\right]$$

where $\sigma_f \ll 1/2$, so that $Q(f)$ is bandlimited as shown in Figure 3.6. Then, if $Q(f) \gg \sigma^2$, we have approximately

$$\text{var}(\hat{f}_c) \geq \frac{1}{N\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{f^2}{\sigma_f^4} df} = \frac{12\sigma_f^4}{N}.$$

Narrower bandwidth (smaller σ_f^2) spectra yield lower bounds for the center frequency since the PSD changes more rapidly as f_c changes. See also Problem 3.16 for another example. ◇

3.11 Signal Processing Examples

We now apply the theory of the CRLB to several signal processing problems of interest. The problems to be considered and some of their areas of application are:

1. Range estimation - sonar, radar, robotics
2. Frequency estimation - sonar, radar, econometrics, spectrometry
3. Bearing estimation - sonar, radar
4. Autoregressive parameter estimation - speech, econometrics.

These examples will be revisited in Chapter 7, in which actual estimators that asymptotically attain the CRLB will be studied.

Example 3.13 - Range Estimation

In radar or active sonar a signal pulse is transmitted. The round trip delay τ_0 from the transmitter to the target and back is related to the range R as $\tau_0 = 2R/c$, where c is the speed of propagation. Estimation of range is therefore equivalent to estimation of the time delay, assuming that c is known. If $s(t)$ is the transmitted signal, a simple model for the received continuous waveform is

$$x(t) = s(t - \tau_0) + w(t) \qquad 0 \leq t \leq T.$$

The transmitted signal pulse is assumed to be nonzero over the interval $[0, T_s]$. Additionally, the signal is assumed to be essentially bandlimited to B Hz. If the maximum time delay is $\tau_{0_{\max}}$, then the observation interval is chosen to include the entire signal by letting $T = T_s + \tau_{0_{\max}}$. The noise is modeled as Gaussian with PSD and ACF as

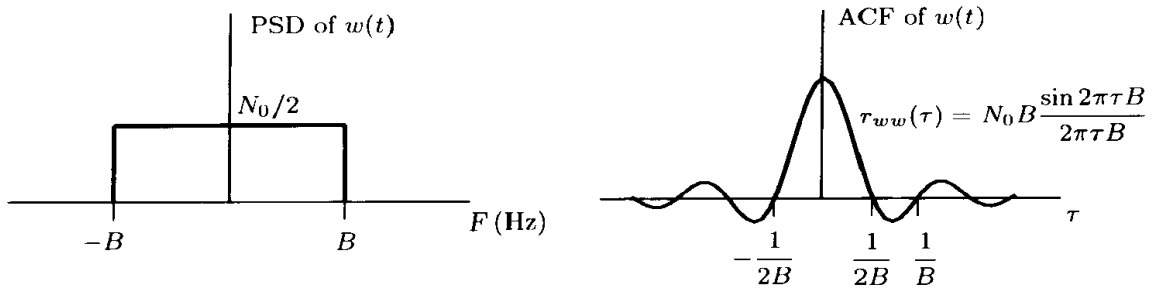


Figure 3.7 Properties of Gaussian observation noise

shown in Figure 3.7. The bandlimited nature of the noise results from filtering the continuous waveform to the signal bandwidth of B Hz. The continuous received waveform is sampled at the Nyquist rate or samples are taken every $\Delta = 1/(2B)$ seconds to form the observed data

$$x(n\Delta) = s(n\Delta - \tau_0) + w(n\Delta) \qquad n = 0, 1, \ldots, N-1.$$

Letting $x[n]$ and $w[n]$ be the sampled sequences, we have our discrete data model

$$x[n] = s(n\Delta - \tau_0) + w[n]. \tag{3.35}$$

Note that $w[n]$ is WGN since the samples are separated by $k\Delta = k/(2B)$, which corresponds to the zeros of the ACF of $w(t)$ as shown in Figure 3.7. Also, $w[n]$ has variance $\sigma^2 = r_{ww}(0) = N_0 B$. Because the signal is nonzero only over the interval $\tau_0 \leq t \leq \tau_0 + T_s$, (3.35) reduces to

$$x[n] = \begin{cases} w[n] & 0 \leq n \leq n_0 - 1 \\ s(n\Delta - \tau_0) + w[n] & n_0 \leq n \leq n_0 + M - 1 \\ w[n] & n_0 + M \leq n \leq N - 1 \end{cases} \tag{3.36}$$

where M is the length of the sampled signal and $n_0 = \tau_0/\Delta$ is the delay in samples. (For simplicity we assume that Δ is so small that τ_0/Δ can be approximated by an integer.) With this formulation we can apply (3.14) in evaluating the CRLB.

$$\begin{aligned} \text{var}(\hat{\tau}_0) &\geq \frac{\sigma^2}{\sum_{n=0}^{N-1} \left(\frac{\partial s[n;\tau_0]}{\partial \tau_0} \right)^2} \\ &= \frac{\sigma^2}{\sum_{n=n_0}^{n_0+M-1} \left(\frac{\partial s(n\Delta - \tau_0)}{\partial \tau_0} \right)^2} \\ &= \frac{\sigma^2}{\sum_{n=n_0}^{n_0+M-1} \left(\frac{ds(t)}{dt}\Big|_{t=n\Delta-\tau_0} \right)^2} \end{aligned}$$

$$= \frac{\sigma^2}{\sum_{n=0}^{M-1} \left(\frac{ds(t)}{dt}\Big|_{t=n\Delta} \right)^2}$$

since $\tau_0 = n_0\Delta$. Assuming that Δ is small enough to approximate the sum by an integral, we have

$$\text{var}(\hat{\tau}_0) \geq \frac{\sigma^2}{\frac{1}{\Delta}\int_0^{T_s} \left(\frac{ds(t)}{dt} \right)^2 dt}.$$

Finally, noting that $\Delta = 1/(2B)$ and $\sigma^2 = N_0 B$, we have

$$\text{var}(\hat{\tau}_0) \geq \frac{\frac{N_0}{2}}{\int_0^{T_s} \left(\frac{ds(t)}{dt} \right)^2 dt}. \tag{3.37}$$

An alternative form observes that the energy $\mathcal{E}$ is

$$\mathcal{E} = \int_0^{T_s} s^2(t)\, dt$$

which results in

$$\text{var}(\hat{\tau}_0) \geq \frac{1}{\frac{\mathcal{E}}{N_0/2}\overline{F^2}} \tag{3.38}$$

where

$$\overline{F^2} = \frac{\int_0^{T_s} \left(\frac{ds(t)}{dt} \right)^2 dt}{\int_0^{T_s} s^2(t) dt}.$$

It can be shown that $\mathcal{E}/(N_0/2)$ is a SNR [Van Trees 1968]. Also, $\overline{F^2}$ is a measure of the bandwidth of the signal since, using standard Fourier transform properties,

$$\overline{F^2} = \frac{\int_{-\infty}^{\infty} (2\pi F)^2 |S(F)|^2 dF}{\int_{-\infty}^{\infty} |S(F)|^2 dF} \tag{3.39}$$

where F denotes continuous-time frequency, and $S(F)$ is the Fourier transform of $s(t)$. In this form it becomes clear that $\overline{F^2}$ is the *mean square bandwidth* of the signal. From (3.38) and (3.39), the larger the mean square bandwidth, the lower the CRLB. For instance, assume that the signal is a Gaussian pulse given by $s(t) = \exp(-\frac{1}{2}\sigma_F^2(t - T_s/2)^2)$ and that $s(t)$ is essentially nonzero over the interval $[0, T_s]$. Then $|S(F)| =$

$(\sigma_F/\sqrt{2\pi})\exp(-2\pi^2F^2/\sigma_F^2)$ and $\overline{F^2} = \sigma_F^2/2$. As the mean square bandwidth increases, the signal pulse becomes narrower and it becomes easier to estimate the time delay.

Finally, by noting that $R = c\tau_0/2$ and using (3.16), the CRLB for range is

$$\text{var}(\hat{R}) \geq \frac{c^2/4}{\frac{\mathcal{E}}{N_0/2}\overline{F^2}}. \tag{3.40}$$

◇

Example 3.14 - Sinusoidal Parameter Estimation

In many fields we are confronted with the problem of estimating the parameters of a sinusoidal signal. Economic data which are cyclical in nature may naturally fit such a model, while in sonar and radar physical mechanisms cause the observed signal to be sinusoidal. Hence, we examine the determination of the CRLB for the amplitude A, frequency f_0, and phase ϕ of a sinusoid embedded in WGN. This example generalizes Examples 3.4 and 3.5. The data are assumed to be

$$x[n] = A\cos(2\pi f_0 n + \phi) + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $A > 0$ and $0 < f_0 < 1/2$ (otherwise the parameters are not identifiable, as is verified by considering $A = 1, \phi = 0$ versus $A = -1, \phi = \pi$ or $f_0 = 0$ with $A = 1/2, \phi = 0$ versus $A = 1/\sqrt{2}, \phi = \pi/4$). Since multiple parameters are unknown, we use (3.33)

$$[\mathbf{I}(\boldsymbol{\theta})]_{ij} = \frac{1}{\sigma^2}\sum_{n=0}^{N-1}\frac{\partial s[n;\boldsymbol{\theta}]}{\partial\theta_i}\frac{\partial s[n;\boldsymbol{\theta}]}{\partial\theta_j}$$

for $\boldsymbol{\theta} = [A\, f_0\, \phi]^T$. In evaluating the CRLB it is assumed that f_0 is not near 0 or 1/2, which allows us to make certain simplifications based on the approximations [Stoica 1989] (see also Problem 3.7):

$$\begin{aligned}\frac{1}{N^{i+1}}\sum_{n=0}^{N-1} n^i \sin(4\pi f_0 n + 2\phi) &\approx 0\\ \frac{1}{N^{i+1}}\sum_{n=0}^{N-1} n^i \cos(4\pi f_0 n + 2\phi) &\approx 0\end{aligned}$$

for $i = 0, 1, 2$. Using these approximations and letting $\alpha = 2\pi f_0 n + \phi$, we have

$$\begin{aligned}[\mathbf{I}(\boldsymbol{\theta})]_{11} &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1}\cos^2\alpha = \frac{1}{\sigma^2}\sum_{n=0}^{N-1}\left(\frac{1}{2}+\frac{1}{2}\cos 2\alpha\right) \approx \frac{N}{2\sigma^2}\\ [\mathbf{I}(\boldsymbol{\theta})]_{12} &= -\frac{1}{\sigma^2}\sum_{n=0}^{N-1} A2\pi n\cos\alpha\sin\alpha = -\frac{\pi A}{\sigma^2}\sum_{n=0}^{N-1} n\sin 2\alpha \approx 0\end{aligned}$$

$$
\begin{aligned}
[\mathbf{I}(\boldsymbol{\theta})]_{13} &= -\frac{1}{\sigma^2}\sum_{n=0}^{N-1} A\cos\alpha\sin\alpha = -\frac{A}{2\sigma^2}\sum_{n=0}^{N-1}\sin 2\alpha \approx 0 \\
[\mathbf{I}(\boldsymbol{\theta})]_{22} &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1} A^2(2\pi n)^2\sin^2\alpha = \frac{(2\pi A)^2}{\sigma^2}\sum_{n=0}^{N-1} n^2\left(\frac{1}{2}-\frac{1}{2}\cos 2\alpha\right) \\
&\approx \frac{(2\pi A)^2}{2\sigma^2}\sum_{n=0}^{N-1} n^2 \\
[\mathbf{I}(\boldsymbol{\theta})]_{23} &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1} A^2 2\pi n\sin^2\alpha \approx \frac{\pi A^2}{\sigma^2}\sum_{n=0}^{N-1} n \\
[\mathbf{I}(\boldsymbol{\theta})]_{33} &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1} A^2\sin^2\alpha \approx \frac{NA^2}{2\sigma^2}.
\end{aligned}
$$

The Fisher information matrix becomes

$$
\mathbf{I}(\boldsymbol{\theta}) = \frac{1}{\sigma^2}\begin{bmatrix}
\frac{N}{2} & 0 & 0 \\
0 & 2A^2\pi^2\sum_{n=0}^{N-1} n^2 & \pi A^2\sum_{n=0}^{N-1} n \\
0 & \pi A^2\sum_{n=0}^{N-1} n & \frac{NA^2}{2}
\end{bmatrix}.
$$

Using (3.22), we have upon inversion

$$
\begin{aligned}
\mathrm{var}(\hat{A}) &\geq \frac{2\sigma^2}{N} \\
\mathrm{var}(\hat{f}_0) &\geq \frac{12}{(2\pi)^2\eta N(N^2-1)} \\
\mathrm{var}(\hat{\phi}) &\geq \frac{2(2N-1)}{\eta N(N+1)}
\end{aligned} \tag{3.41}
$$

where $\eta = A^2/(2\sigma^2)$ is the SNR. Frequency estimation of a sinusoid is of considerable interest. Note that the CRLB for the frequency decreases as the SNR increases and that the bound decreases as $1/N^3$, making it quite sensitive to data record length. See also Problem 3.17 for a variation of this example.

◇

Example 3.15 - Bearing Estimation

In sonar it is of interest to estimate the bearing to a target as shown in Figure 3.8. To do so the acoustic pressure field is observed by an array of equally spaced sensors in a

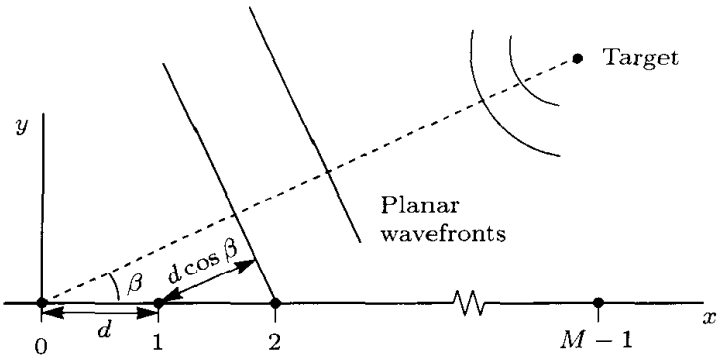


Figure 3.8 Geometry of array for bearing estimation

line. Assuming that the target radiates a sinusoidal signal $A\cos(2\pi F_0 t + \phi)$, then the received signal at the nth sensor is $A\cos(2\pi F_0(t - t_n) + \phi)$, where t_n is the propagation time to the nth sensor. If the array is located far from the target, then the circular wavefronts can be considered to be planar at the array. As shown in Figure 3.8, the wavefront at the $(n-1)$st sensor lags that at the nth sensor by $d\cos\beta/c$ due to the extra propagation distance. Thus, the propagation time to the nth sensor is

$$t_n = t_0 - n\frac{d}{c}\cos\beta \qquad n = 0, 1, \ldots, M-1$$

where t_0 is the propagation time to the zeroth sensor, and the observed signal at the nth sensor is

$$s_n(t) = A\cos\left[2\pi F_0(t - t_0 + n\frac{d}{c}\cos\beta) + \phi\right].$$

If a single "snapshot" of data is taken or the array element outputs are sampled at a given time t_s, then

$$s_n(t_s) = A\cos[2\pi(F_0\frac{d}{c}\cos\beta)n + \phi'] \tag{3.42}$$

where $\phi' = \phi + 2\pi F_0(t_s - t_0)$. In this form it becomes clear that the spatial observations are sinusoidal with frequency $f_s = F_0(d/c)\cos\beta$. To complete the description of the data we assume that the sensor outputs are corrupted by Gaussian noise with zero mean and variance σ^2 which is independent from sensor to sensor. The data are modeled as

$$x[n] = s_n(t_s) + w[n] \qquad n = 0, 1, \ldots, M-1$$

where $w[n]$ is WGN. Since typically A, ϕ are unknown, as well as β, we have the problem of estimating $\{A, f_s, \phi'\}$ based on (3.42) as in Example 3.14. Once the CRLB for these parameters is determined, we can use the transformation of parameters formula. The transformation is for $\boldsymbol{\theta} = [A\ f_s\ \phi']^T$

$$\boldsymbol{\alpha} = \mathbf{g}(\boldsymbol{\theta}) = \begin{bmatrix} A \\ \beta \\ \phi' \end{bmatrix} = \begin{bmatrix} A \\ \arccos\left(\frac{cf_s}{F_0 d}\right) \\ \phi' \end{bmatrix}.$$

The Jacobian is

$$\frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{c}{F_0 d \sin\beta} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

so that from (3.30)

$$\left[\mathbf{C}_{\hat{\alpha}} - \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{g}(\boldsymbol{\theta})^T}{\partial \boldsymbol{\theta}}\right]_{22} \geq 0.$$

Because of the diagonal Jacobian this yields

$$\mathrm{var}(\hat{\beta}) \geq \left[\frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right]_{22}^2 \left[\mathbf{I}^{-1}(\boldsymbol{\theta})\right]_{22}.$$

But from (3.41) we have

$$\left[\mathbf{I}^{-1}(\boldsymbol{\theta})\right]_{22} = \frac{12}{(2\pi)^2 \eta M(M^2-1)}$$

and therefore

$$\mathrm{var}(\hat{\beta}) \geq \frac{12}{(2\pi)^2 \eta M(M^2-1)} \frac{c^2}{F_0^2 d^2 \sin^2\beta}$$

or finally

$$\mathrm{var}(\hat{\beta}) \geq \frac{12}{(2\pi)^2 M \eta \frac{M+1}{M-1} \left(\frac{L}{\lambda}\right)^2 \sin^2\beta} \tag{3.43}$$

where $\lambda = c/F_0$ is the wavelength of the propagating plane wave and $L = (M-1)d$ is the length of the array. Note that it is easiest to estimate bearing if $\beta = 90°$, and impossible if $\beta = 0°$. Also, the bound depends critically on the array length in wavelengths or L/λ, as well as the SNR at the array output or $M\eta$. The use of the CRLB for feasibility studies is examined in Problem 3.19. ◇

Example 3.16 - Autoregressive Parameter Estimation

In speech processing an important model for speech production is the autoregressive (AR) process. As shown in Figure 3.9, the data are modeled as the output of a *causal all-pole discrete filter excited at the input by WGN* $u[n]$. The excitation noise $u[n]$ is an inherent part of the model, necessary to ensure that $x[n]$ is a WSS random process. The all-pole filter acts to model the vocal tract, while the excitation noise models the forcing of air through a constriction in the throat necessary to produce an unvoiced sound such as an "s." The effect of the filter is to color the white noise so as to model PSDs with several resonances. This model is also referred to as a *linear predictive coding* (LPC) model [Makhoul 1975]. Since the AR model is capable of producing a variety of PSDs, depending on the choice of the AR filter parameters $\{a[1], a[2], \ldots, a[p]\}$ and

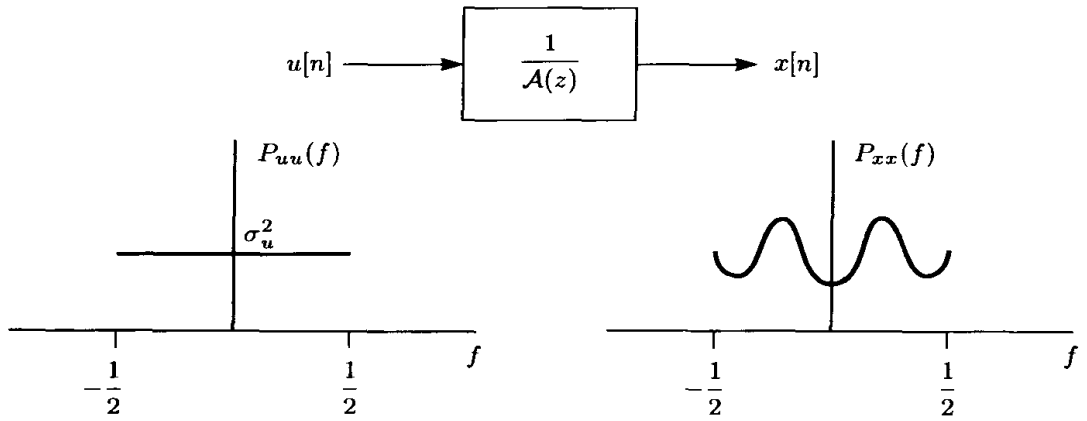


Figure 3.9 Autoregressive model

$$\mathcal{A}(z) = 1 + \sum_{m=1}^{p} a[m]z^{-m}$$

excitation white noise variance σ_u^2, it has also been successfully used for high-resolution spectral estimation. Based on observed data $\{x[0], x[1], \ldots, x[N-1]\}$, the parameters are estimated (see Example 7.18), and hence the PSD is estimated as [Kay 1988]

$$\hat{P}_{xx}(f) = \frac{\hat{\sigma_u^2}}{\left|1 + \sum_{m=1}^{p} \hat{a}[m] \exp(-j2\pi fm)\right|^2}.$$

The derivation of the CRLB proves to be a difficult task. The interested reader may consult [Box and Jenkins 1970, Porat and Friedlander 1987] for details. In practice the asymptotic CRLB given by (3.34) is quite accurate, even for short data records of about $N = 100$ points if the poles are not too close to the unit circle in the z plane. Therefore, we now determine the asymptotic CRLB. The PSD implied by the AR model is

$$P_{xx}(f; \boldsymbol{\theta}) = \frac{\sigma_u^2}{|A(f)|^2}$$

where $\boldsymbol{\theta} = [a[1]\, a[2] \ldots a[p]\, \sigma_u^2]^T$ and $A(f) = 1 + \sum_{m=1}^{p} a[m] \exp(-j2\pi fm)$. The partial derivatives are

$$\begin{aligned}
\frac{\partial \ln P_{xx}(f; \boldsymbol{\theta})}{\partial a[k]} &= -\frac{\partial \ln |A(f)|^2}{\partial a[k]} \\
&= -\frac{1}{|A(f)|^2} \left[A(f) \exp(j2\pi fk) + A^*(f) \exp(-j2\pi fk)\right] \\
\frac{\partial \ln P_{xx}(f; \boldsymbol{\theta})}{\partial \sigma_u^2} &= \frac{1}{\sigma_u^2}.
\end{aligned}$$

For $k = 1, 2, \ldots, p$; $l = 1, 2, \ldots, p$ we have from (3.34)

$$\begin{aligned}[\mathbf{I}(\boldsymbol{\theta})]_{kl} &= \frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{|A(f)|^4}\left[A(f)\exp(j2\pi fk) + A^*(f)\exp(-j2\pi fk)\right] \\ &\qquad \cdot \left[A(f)\exp(j2\pi fl) + A^*(f)\exp(-j2\pi fl)\right] df \\ &= \frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \left[\frac{1}{A^*(f)^2}\exp[j2\pi f(k+l)] + \frac{1}{|A(f)|^2}\exp[j2\pi f(k-l)]\right. \\ &\quad \left. + \frac{1}{|A(f)|^2}\exp[j2\pi f(l-k)] + \frac{1}{A^2(f)}\exp[-j2\pi f(k+l)]\right] df.\end{aligned}$$

Noting that

$$\begin{aligned}\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{A^*(f)^2}\exp[j2\pi f(k+l)]\,df &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{A^2(f)}\exp[-j2\pi f(k+l)]\,df \\ \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{|A(f)|^2}\exp[j2\pi f(k-l)]\,df &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{|A(f)|^2}\exp[j2\pi f(l-k)]\,df\end{aligned}$$

which follows from the Hermitian property of the integrand (due to $A(-f) = A^*(f)$), we have

$$\begin{aligned}[\mathbf{I}(\boldsymbol{\theta})]_{kl} &= N\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{|A(f)|^2}\exp[j2\pi f(k-l)]\,df \\ &\quad + N\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{A^*(f)^2}\exp[j2\pi f(k+l)]\,df.\end{aligned}$$

The second integral is the inverse Fourier transform of $1/A^*(f)^2$ evaluated at $n = k+l > 0$. This term is zero since the sequence is the convolution of two anticausal sequences, that is, if

$$\mathcal{F}^{-1}\left\{\frac{1}{A(f)}\right\} = \begin{cases} h[n] & n \geq 0 \\ 0 & n < 0 \end{cases}$$

then

$$\begin{aligned}\mathcal{F}^{-1}\left\{\frac{1}{A^*(f)^2}\right\} &= h[-n] \star h[-n] \\ &= 0 \qquad \text{for } n > 0.\end{aligned}$$

Therefore,

$$[\mathbf{I}(\boldsymbol{\theta})]_{kl} = \frac{N}{\sigma_u^2} r_{xx}[k-l].$$

For $k = 1, 2, \ldots, p$; $l = p + 1$

$$\begin{aligned}
[\mathbf{I}(\boldsymbol{\theta})]_{kl} &= -\frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\sigma_u^2}\frac{1}{|A(f)|^2}\left[A(f)\exp(j2\pi fk) + A^*(f)\exp(-j2\pi fk)\right] df \\
&= -\frac{N}{\sigma_u^2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{A^*(f)}\exp(j2\pi fk)\, df \\
&= 0
\end{aligned}$$

where again we have used the Hermitian property of the integrand and the anticausality of $\mathcal{F}^{-1}\{1/A^*(f)\}$. Finally, for $k = p + 1; l = p + 1$

$$[\mathbf{I}(\boldsymbol{\theta})]_{kl} = \frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\sigma_u^4}\, df = \frac{N}{2\sigma_u^4}$$

so that

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} \frac{N}{\sigma_u^2}\mathbf{R}_{xx} & \mathbf{0} \\ \mathbf{0}^T & \frac{N}{2\sigma_u^4} \end{bmatrix} \tag{3.44}$$

where $[\mathbf{R}_{xx}]_{ij} = r_{xx}[i-j]$ is a $p \times p$ Toeplitz autocorrelation matrix and $\mathbf{0}$ is a $p \times 1$ vector of zeros. Upon inverting the Fisher information matrix we have that

$$\begin{aligned}
\mathrm{var}(\hat{a}[k]) &\geq \frac{\sigma_u^2}{N}\left[\mathbf{R}_{xx}^{-1}\right]_{kk} \qquad k = 1, 2, \ldots, p \\
\mathrm{var}(\hat{\sigma_u^2}) &\geq \frac{2\sigma_u^4}{N}.
\end{aligned} \tag{3.45}$$

As an illustration, if $p = 1$,

$$\mathrm{var}(\hat{a}[1]) \geq \frac{\sigma_u^2}{N r_{xx}[0]}.$$

But

$$r_{xx}[0] = \frac{\sigma_u^2}{1 - a^2[1]}$$

so that

$$\mathrm{var}(\hat{a}[1]) \geq \frac{1}{N}\left(1 - a^2[1]\right)$$

indicating that it is easier to estimate the filter parameter when $|a[1]|$ is closer to one than to zero. Since the pole of the filter is at $-a[1]$, this means that the filter parameters of processes with PSDs having sharp peaks are more easily estimated (see also Problem 3.20).

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Problems

3.1 If $x[n]$ for $n = 0, 1, \ldots, N-1$ are IID according to $\mathcal{U}[0, \theta]$, show that the regularity condition does not hold or that

$$E\left[\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right] \neq 0 \qquad \text{for all } \theta > 0.$$

Hence, the CRLB cannot be applied to this problem.

3.2 In Example 3.1 assume that $w[0]$ has the PDF $p(w[0])$ which can now be arbitrary. Show that the CRLB for A is

$$\text{var}(\hat{A}) \geq \left[\int_{-\infty}^{\infty} \frac{\left(\frac{dp(u)}{du}\right)^2}{p(u)} du\right]^{-1}.$$

Evaluate this for the Laplacian PDF

$$p(w[0]) = \frac{1}{\sqrt{2}\sigma} \exp\left(-\frac{\sqrt{2}|w[0]|}{\sigma}\right)$$

and compare the result to the Gaussian case.

3.3 The data $x[n] = Ar^n + w[n]$ for $n = 0, 1, \ldots, N-1$ are observed, where $w[n]$ is WGN with variance σ^2 and $r > 0$ is known. Find the CRLB for A. Show that an efficient estimator exists and find its variance. What happens to the variance as $N \to \infty$ for various values of r?

3.4 If $x[n] = r^n + w[n]$ for $n = 0, 1, \ldots, N-1$ are observed, where $w[n]$ is WGN with variance σ^2 and r is to be estimated, find the CRLB. Does an efficient estimator exist and if so find its variance?

3.5 If $x[n] = A + w[n]$ for $n = 0, 1, \ldots, N-1$ are observed and $\mathbf{w} = [w[0]\, w[1] \ldots w[N-1]]^T \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$, find the CRLB for A. Does an efficient estimator exist and if so what is its variance?

3.6 For Example 2.3 compute the CRLB. Does it agree with the results given?

3.7 Prove that in Example 3.4

$$\frac{1}{N} \sum_{n=0}^{N-1} \cos(4\pi f_0 n + 2\phi) \approx 0.$$

What conditions on f_0 are required for this to hold? Hint: Note that

$$\sum_{n=0}^{N-1} \cos(\alpha n + \beta) = \text{Re}\left(\sum_{n=0}^{N-1} \exp[j(\alpha n + \beta)] \right)$$

and use the geometric progression sum formula.

3.8 Repeat the computation of the CRLB for Example 3.3 by using the alternative expression (3.12).

3.9 We observe two samples of a DC level in *correlated* Gaussian noise

$$\begin{aligned} x[0] &= A + w[0] \\ x[1] &= A + w[1] \end{aligned}$$

where $\mathbf{w} = [w[0]\, w[1]]^T$ is zero mean with covariance matrix

$$\mathbf{C} = \sigma^2 \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

The parameter ρ is the correlation coefficient between $w[0]$ and $w[1]$. Compute the CRLB for A and compare it to the case when $w[n]$ is WGN or $\rho = 0$. Also, explain what happens when $\rho \to \pm 1$. Finally, comment on the additivity property of the Fisher information for nonindependent observations.

3.10 By using (3.23) prove that the Fisher information matrix is positive semidefinite for all $\boldsymbol{\theta}$. In practice, we assume it to be positive definite and hence invertible, although this is not always the case. Consider the data model in Problem 3.3 with the modification that r is unknown. Find the Fisher information matrix for $\boldsymbol{\theta} = [A\ r]^T$. Are there any values of $\boldsymbol{\theta}$ for which $\mathbf{I}(\boldsymbol{\theta})$ is not positive definite?

3.11 For a 2×2 Fisher information matrix

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

which is positive definite, show that

$$[\mathbf{I}^{-1}(\boldsymbol{\theta})]_{11} = \frac{c}{ac - b^2} \geq \frac{1}{a} = \frac{1}{[\mathbf{I}(\boldsymbol{\theta})]_{11}}.$$

What does this say about estimating a parameter when a second parameter is either known or unknown? When does equality hold and why?

3.12 Prove that

$$[\mathbf{I}^{-1}(\boldsymbol{\theta})]_{ii} \geq \frac{1}{[\mathbf{I}(\boldsymbol{\theta})]_{ii}}.$$

This generalizes the result of Problem 3.11. Additionally, it provides another lower bound on the variance, although it is usually not attainable. Under what conditions will the new bound be achieved? Hint: Apply the Cauchy-Schwarz inequality to $\mathbf{e}_i^T \sqrt{\mathbf{I}(\boldsymbol{\theta})} \sqrt{\mathbf{I}^{-1}(\boldsymbol{\theta})} \mathbf{e}_i$, where $\mathbf{e}_i$ is the vectors of all zeros except for a 1 as the ith element. The square root of a positive definite matrix $\mathbf{A}$ is defined to be the matrix with the same eigenvectors as $\mathbf{A}$ but whose eigenvalues are the square roots of those of $\mathbf{A}$.

3.13 Consider a generalization of the line fitting problem as described in Example 3.7, termed *polynomial or curve fitting.* The data model is

$$x[n] = \sum_{k=0}^{p-1} A_k n^k + w[n]$$

for $n = 0, 1, \ldots, N-1$. As before, $w[n]$ is WGN with variance σ^2. It is desired to estimate $\{A_0, A_1, \ldots, A_{p-1}\}$. Find the Fisher information matrix for this problem.

3.14 For the data model in Example 3.11 consider the estimator $\hat{\sigma_A^2} = (\hat{A})^2$, where $\hat{A}$ is the sample mean. Assume we observe a given data set in which the realization of the random variable A is the value A_0. Show that $\hat{A} \to A_0$ as $N \to \infty$ by verifying that

$$\begin{aligned} E(\hat{A}|A = A_0) &= A_0 \\ \text{var}(\hat{A}|A = A_0) &= \frac{\sigma^2}{N}. \end{aligned}$$

Hence, $\hat{\sigma_A^2} \to A_0^2$ as $N \to \infty$ for the given realization $A = A_0$. Next find the variance of $\hat{\sigma_A^2}$ as $N \to \infty$ by determining $\text{var}(A^2)$, where $A \sim \mathcal{N}(0, \sigma_A^2)$, and compare it to the CRLB. Explain why σ_A^2 cannot be estimated without error even for $N \to \infty$.

3.15 Independent bivariate Gaussian samples $\{\mathbf{x}[0], \mathbf{x}[1], \ldots, \mathbf{x}[N-1]\}$ are observed. Each observation is a 2×1 vector which is distributed as $\mathbf{x}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$ and

$$\mathbf{C} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

Find the CRLB for the correlation coefficient ρ. Hint: Use (3.32).

3.16 It is desired to estimate the total power P_0 of a WSS random process, whose PSD is given as

$$P_{xx}(f) = P_0 Q(f)$$

where

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} Q(f)\, df = 1$$

and $Q(f)$ is known. If N observations are available, find the CRLB for the total power using the exact form (3.32) as well as the asymptotic approximation (3.34) and compare.

3.17 If in Example 3.14 the data are observed over the interval $n = -M, \ldots, 0, \ldots, M$, find the Fisher information matrix. What is the CRLB for the sinusoidal parameters? You can use the same approximations as in the example by assuming M is large. Compare the results to that of the example.

3.18 Using the results of Example 3.13, determine the best range estimation accuracy of a sonar if

$$s(t) = \begin{cases} 1 - 100|t - 0.01| & 0 \le t \le 0.02 \\ 0 & \text{otherwise.} \end{cases}$$

Let $N_0/2 = 10^{-6}$ and $c = 1500$ m/s.

3.19 A line array of antennas with $d = \lambda/2$ spacing for a 1 MHz electromagnetic signal is to be designed to yield a 5° standard deviation at $\beta = 90^\circ$. If the SNR η is 0 dB, comment on the feasibility of the requirement. Assume that the line array is to be mounted on the wing of an aircraft. Use $c = 3 \times 10^8$ m/s.

3.20 In Example 3.16 we found the CRLB for the filter parameter of an AR(1) process. Assuming σ_u^2 is known, find the asymptotic CRLB for $P_{xx}(f)$ at a given frequency by using (3.16) and (3.44). For $a[1] = -0.9$, $\sigma_u^2 = 1$, and $N = 100$, plot the CRLB versus frequency. Explain your results.

Appendix 3A

Derivation of Scalar Parameter CRLB

In this appendix we derive the CRLB for a scalar parameter $\alpha = g(\theta)$ where the PDF is parameterized by θ. We consider all unbiased estimators $\hat{\alpha}$ or those for which

$$E(\hat{\alpha}) = \alpha = g(\theta)$$

or

$$\int \hat{\alpha} p(\mathbf{x}; \theta)\, d\mathbf{x} = g(\theta). \tag{3A.1}$$

Before beginning the derivation we first examine the regularity condition

$$E\left[\frac{\partial \ln p(\mathbf{x}; \theta)}{\partial \theta}\right] = 0, \tag{3A.2}$$

which is assumed to hold. Note that

$$\begin{aligned}
\int \frac{\partial \ln p(\mathbf{x}; \theta)}{\partial \theta} p(\mathbf{x}; \theta)\, d\mathbf{x} &= \int \frac{\partial p(\mathbf{x}; \theta)}{\partial \theta}\, d\mathbf{x} \\
&= \frac{\partial}{\partial \theta} \int p(\mathbf{x}; \theta)\, d\mathbf{x} \\
&= \frac{\partial 1}{\partial \theta} \\
&= 0.
\end{aligned}$$

Hence, we conclude that the regularity condition will be satisfied if the order of differentiation and integration may be interchanged. This is generally true except when the domain of the PDF for which it is nonzero depends on the unknown parameter such as in Problem 3.1.

Now differentiating both sides of (3A.1) with respect to θ and interchanging the partial differentiation and integration produces

$$\int \hat{\alpha} \frac{\partial p(\mathbf{x}; \theta)}{\partial \theta}\, d\mathbf{x} = \frac{\partial g(\theta)}{\partial \theta}$$

or

$$\int \hat{\alpha} \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta)\, d\mathbf{x} = \frac{\partial g(\theta)}{\partial \theta}. \tag{3A.3}$$

We can modify this using the regularity condition to produce

$$\int (\hat{\alpha} - \alpha) \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta)\, d\mathbf{x} = \frac{\partial g(\theta)}{\partial \theta} \tag{3A.4}$$

since

$$\int \alpha \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta)\, d\mathbf{x} = \alpha E\left[\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right] = 0.$$

We now apply the Cauchy-Schwarz inequality

$$\left[\int w(\mathbf{x}) g(\mathbf{x}) h(\mathbf{x})\, d\mathbf{x}\right]^2 \leq \int w(\mathbf{x}) g^2(\mathbf{x})\, d\mathbf{x} \int w(\mathbf{x}) h^2(\mathbf{x})\, d\mathbf{x} \tag{3A.5}$$

which holds with equality if and only if $g(\mathbf{x}) = ch(\mathbf{x})$ for c some constant not dependent on $\mathbf{x}$. The functions g and h are arbitrary scalar functions, while $w(\mathbf{x}) \geq 0$ for all $\mathbf{x}$. Now let

$$\begin{aligned} w(\mathbf{x}) &= p(\mathbf{x};\theta) \\ g(\mathbf{x}) &= \hat{\alpha} - \alpha \\ h(\mathbf{x}) &= \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} \end{aligned}$$

and apply the Cauchy-Schwarz inequality to (3A.4) to produce

$$\left(\frac{\partial g(\theta)}{\partial \theta}\right)^2 \leq \int (\hat{\alpha} - \alpha)^2 p(\mathbf{x};\theta)\, d\mathbf{x} \int \left(\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right)^2 p(\mathbf{x};\theta)\, d\mathbf{x}$$

or

$$\mathrm{var}(\hat{\alpha}) \geq \frac{\left(\frac{\partial g(\theta)}{\partial \theta}\right)^2}{E\left[\left(\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right)^2\right]}.$$

Now note that

$$E\left[\left(\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right)^2\right] = -E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right].$$

This follows from (3A.2) as

$$\begin{aligned} E\left[\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right] &= 0 \\ \int \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta)\, d\mathbf{x} &= 0 \end{aligned}$$

$$\frac{\partial}{\partial\theta}\int \frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta} p(\mathbf{x};\theta)\,d\mathbf{x} = 0$$

$$\int \left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2} p(\mathbf{x};\theta) + \frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\frac{\partial p(\mathbf{x};\theta)}{\partial\theta}\right] d\mathbf{x} = 0$$

or

$$-E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right] = \int \frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta} p(\mathbf{x};\theta)\,d\mathbf{x}$$

$$= E\left[\left(\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\right)^2\right].$$

Thus,

$$\mathrm{var}(\hat{\alpha}) \geq \frac{\left(\frac{\partial g(\theta)}{\partial\theta}\right)^2}{-E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right]}$$

which is (3.16). If $\alpha = g(\theta) = \theta$, we have (3.6).

Note that the condition for equality is

$$\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta} = \frac{1}{c}(\hat{\alpha} - \alpha)$$

where c can depend on θ but not on $\mathbf{x}$. If $\alpha = g(\theta) = \theta$, we have

$$\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta} = \frac{1}{c(\theta)}(\hat{\theta} - \theta).$$

The possible dependence of c on θ is noted. To determine $c(\theta)$

$$\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2} = -\frac{1}{c(\theta)} + \frac{\partial\left(\frac{1}{c(\theta)}\right)}{\partial\theta}(\hat{\theta} - \theta)$$

$$-E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right] = \frac{1}{c(\theta)}$$

or finally

$$c(\theta) = \frac{1}{-E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right]}$$

$$= \frac{1}{I(\theta)}$$

which agrees with (3.7).

Appendix 3B

Derivation of Vector Parameter CRLB

In this appendix the CRLB for a vector parameter $\boldsymbol{\alpha} = \mathbf{g}(\boldsymbol{\theta})$ is derived. The PDF is characterized by $\boldsymbol{\theta}$. We consider unbiased estimators such that

$$E(\hat{\alpha}_i) = \alpha_i = [\mathbf{g}(\boldsymbol{\theta})]_i \qquad i = 1, 2, \ldots, r.$$

The regularity conditions are

$$E\left[\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right] = \mathbf{0}$$

so that

$$\int (\hat{\alpha}_i - \alpha_i) \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_i} p(\mathbf{x};\boldsymbol{\theta})\, d\mathbf{x} = \frac{\partial [\mathbf{g}(\boldsymbol{\theta})]_i}{\partial \theta_i}. \tag{3B.1}$$

Now consider for $j \neq i$

$$\begin{aligned}
\int (\hat{\alpha}_i - \alpha_i) \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_j} p(\mathbf{x};\boldsymbol{\theta})\, d\mathbf{x} &= \int (\hat{\alpha}_i - \alpha_i) \frac{\partial p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_j}\, d\mathbf{x} \\
&= \frac{\partial}{\partial \theta_j} \int \hat{\alpha}_i p(\mathbf{x};\boldsymbol{\theta})\, d\mathbf{x} \\
&\quad - \alpha_i E\left[\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_j}\right] \\
&= \frac{\partial \alpha_i}{\partial \theta_j} \\
&= \frac{\partial [\mathbf{g}(\boldsymbol{\theta})]_i}{\partial \theta_j}.
\end{aligned} \tag{3B.2}$$

Combining (3B.1) and (3B.2) into matrix form, we have

$$\int (\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}) \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T p(\mathbf{x};\boldsymbol{\theta})\, d\mathbf{x} = \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}.$$

Now premultiply by $\mathbf{a}^T$ and postmultiply by $\mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are arbitrary $r \times 1$ and $p \times 1$ vectors, respectively, to yield

$$\int \mathbf{a}^T(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}) \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T \mathbf{b} p(\mathbf{x};\boldsymbol{\theta})\, d\mathbf{x} = \mathbf{a}^T \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{b}.$$

Now let

$$\begin{aligned} w(\mathbf{x}) &= p(\mathbf{x};\boldsymbol{\theta}) \\ g(\mathbf{x}) &= \mathbf{a}^T(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}) \\ h(\mathbf{x}) &= \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T \mathbf{b} \end{aligned}$$

and apply the Cauchy-Schwarz inequality of (3A.5)

$$\begin{aligned} \left(\mathbf{a}^T \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{b}\right)^2 &\le \int \mathbf{a}^T(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha})(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha})^T \mathbf{a} p(\mathbf{x};\boldsymbol{\theta})\, d\mathbf{x} \\ &\quad \cdot \int \mathbf{b}^T \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T \mathbf{b} p(\mathbf{x};\boldsymbol{\theta})\, d\mathbf{x} \\ &= \mathbf{a}^T \mathbf{C}_{\hat{\alpha}} \mathbf{a} \mathbf{b}^T \mathbf{I}(\boldsymbol{\theta}) \mathbf{b} \end{aligned}$$

since as in the scalar case

$$E\left[\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_i} \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_j}\right] = -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j}\right] = [\mathbf{I}(\boldsymbol{\theta})]_{ij}.$$

Since $\mathbf{b}$ was arbitrary, let

$$\mathbf{b} = \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T \mathbf{a}$$

to yield

$$\left(\mathbf{a}^T \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T \mathbf{a}\right)^2 \le \mathbf{a}^T \mathbf{C}_{\hat{\alpha}} \mathbf{a} \left(\mathbf{a}^T \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T \mathbf{a}\right).$$

Since $\mathbf{I}(\boldsymbol{\theta})$ is positive definite, so is $\mathbf{I}^{-1}(\boldsymbol{\theta})$, and $\frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T$ is at least positive semidefinite. The term inside the parentheses is therefore nonnegative, and we have

$$\mathbf{a}^T \left(\mathbf{C}_{\hat{\alpha}} - \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T\right) \mathbf{a} \ge 0.$$

Recall that $\mathbf{a}$ was arbitrary, so that (3.30) follows. If $\boldsymbol{\alpha} = \mathbf{g}(\boldsymbol{\theta}) = \boldsymbol{\theta}$, then $\frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{I}$ and (3.24) follows. The conditions for equality are $g(\mathbf{x}) = ch(\mathbf{x})$, where c is a constant not dependent on $\mathbf{x}$. This condition becomes

$$\begin{aligned} \mathbf{a}^T(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}) &= c \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T \mathbf{b} \\ &= c \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}^T \mathbf{a}. \end{aligned}$$

Since **a** was arbitrary

$$\frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{1}{c}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}).$$

Consider the case when $\boldsymbol{\alpha} = \mathbf{g}(\boldsymbol{\theta}) = \boldsymbol{\theta}$, so that $\frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{I}$. Then,

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{1}{c}\mathbf{I}(\boldsymbol{\theta})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}).$$

Noting that c may depend on $\boldsymbol{\theta}$, we have

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_i} = \sum_{k=1}^{p} \frac{[\mathbf{I}(\boldsymbol{\theta})]_{ik}}{c(\boldsymbol{\theta})}(\hat{\theta}_k - \theta_k)$$

and differentiating once more

$$\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j} = \sum_{k=1}^{p} \left(\frac{[\mathbf{I}(\boldsymbol{\theta})]_{ik}}{c(\boldsymbol{\theta})}(-\delta_{kj}) + \frac{\partial \left(\frac{[\mathbf{I}(\boldsymbol{\theta})]_{ik}}{c(\boldsymbol{\theta})} \right)}{\partial \theta_j}(\hat{\theta}_k - \theta_k) \right).$$

Finally, we have

$$\begin{aligned} [\mathbf{I}(\boldsymbol{\theta})]_{ij} &= -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j}\right] \\ &= \frac{[\mathbf{I}(\boldsymbol{\theta})]_{ij}}{c(\boldsymbol{\theta})} \end{aligned}$$

since $E(\hat{\theta}_k) = \theta_k$. Clearly, $c(\boldsymbol{\theta}) = 1$ and the condition for equality follows.

Appendix 3C

Derivation of General Gaussian CRLB

Assume that $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}(\boldsymbol{\theta}), \mathbf{C}(\boldsymbol{\theta}))$, where $\boldsymbol{\mu}(\boldsymbol{\theta})$ is the $N \times 1$ mean vector and $\mathbf{C}(\boldsymbol{\theta})$ is the $N \times N$ covariance matrix, both of which depend on $\boldsymbol{\theta}$. Then the PDF is

$$p(\mathbf{x};\boldsymbol{\theta}) = \frac{1}{(2\pi)^{\frac{N}{2}} \det^{\frac{1}{2}}[\mathbf{C}(\boldsymbol{\theta})]} \exp\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}(\boldsymbol{\theta}))^T \mathbf{C}^{-1}(\boldsymbol{\theta})(\mathbf{x}-\boldsymbol{\mu}(\boldsymbol{\theta}))\right].$$

We will make use of the following identities

$$\frac{\partial \ln \det[\mathbf{C}(\boldsymbol{\theta})]}{\partial \theta_k} = \mathrm{tr}\left(\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k}\right) \tag{3C.1}$$

where $\partial \mathbf{C}(\boldsymbol{\theta})/\partial\theta_k$ is the $N \times N$ matrix with $[i,j]$ element $\partial[\mathbf{C}(\boldsymbol{\theta})]_{ij}/\partial\theta_k$ and

$$\frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_k} = -\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k}\mathbf{C}^{-1}(\boldsymbol{\theta}). \tag{3C.2}$$

To establish (3C.1) we first note

$$\frac{\partial \ln \det[\mathbf{C}(\boldsymbol{\theta})]}{\partial \theta_k} = \frac{1}{\det[\mathbf{C}(\boldsymbol{\theta})]}\frac{\partial \det[\mathbf{C}(\boldsymbol{\theta})]}{\partial \theta_k}. \tag{3C.3}$$

Since $\det[\mathbf{C}(\boldsymbol{\theta})]$ depends on all the elements of $\mathbf{C}(\boldsymbol{\theta})$

$$\begin{aligned}\frac{\partial \det[\mathbf{C}(\boldsymbol{\theta})]}{\partial \theta_k} &= \sum_{i=1}^{N}\sum_{j=1}^{N}\frac{\partial \det[\mathbf{C}(\boldsymbol{\theta})]}{\partial[\mathbf{C}(\boldsymbol{\theta})]_{ij}}\frac{\partial[\mathbf{C}(\boldsymbol{\theta})]_{ij}}{\partial \theta_k} \\ &= \mathrm{tr}\left(\frac{\partial \det[\mathbf{C}(\boldsymbol{\theta})]}{\partial \mathbf{C}(\boldsymbol{\theta})}\frac{\partial \mathbf{C}^T(\boldsymbol{\theta})}{\partial \theta_k}\right)\end{aligned} \tag{3C.4}$$

where $\partial \det[\mathbf{C}(\boldsymbol{\theta})]/\partial \mathbf{C}(\boldsymbol{\theta})$ is an $N \times N$ matrix with $[i,j]$ element $\partial \det[\mathbf{C}(\boldsymbol{\theta})]/\partial[\mathbf{C}(\boldsymbol{\theta})]_{ij}$ and the identity

$$\mathrm{tr}(\mathbf{A}\mathbf{B}^T) = \sum_{i=1}^{N}\sum_{j=1}^{N}[\mathbf{A}]_{ij}[\mathbf{B}]_{ij}$$

has been used. Now by the definition of the determinant

$$\det[\mathbf{C}(\boldsymbol{\theta})] = \sum_{i=1}^{N} [\mathbf{C}(\boldsymbol{\theta})]_{ij}[\mathbf{M}]_{ij}$$

where $\mathbf{M}$ is the $N \times N$ cofactor matrix and j can take on any value from 1 to N. Thus,

$$\frac{\partial \det[\mathbf{C}(\boldsymbol{\theta})]}{\partial [\mathbf{C}(\boldsymbol{\theta})]_{ij}} = [\mathbf{M}]_{ij}$$

or

$$\frac{\partial \det[\mathbf{C}(\boldsymbol{\theta})]}{\partial \mathbf{C}(\boldsymbol{\theta})} = \mathbf{M}.$$

It is well known, however, that

$$\mathbf{C}^{-1}(\boldsymbol{\theta}) = \frac{\mathbf{M}^T}{\det[\mathbf{C}(\boldsymbol{\theta})]}$$

so that

$$\frac{\partial \det[\mathbf{C}(\boldsymbol{\theta})]}{\partial \mathbf{C}(\boldsymbol{\theta})} = \mathbf{C}^{-1}(\boldsymbol{\theta}) \det[\mathbf{C}(\boldsymbol{\theta})].$$

Using this in (3C.3) and (3C.4), we have the desired result.

$$\begin{aligned} \frac{\partial \ln \det[\mathbf{C}(\boldsymbol{\theta})]}{\partial \theta_k} &= \frac{1}{\det[\mathbf{C}(\boldsymbol{\theta})]} \mathrm{tr}\left(\mathbf{C}^{-1}(\boldsymbol{\theta}) \det[\mathbf{C}(\boldsymbol{\theta})] \frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k} \right) \\ &= \mathrm{tr}\left(\mathbf{C}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k} \right). \end{aligned}$$

The second identity (3C.2) is easily established as follows. Consider

$$\mathbf{C}^{-1}(\boldsymbol{\theta})\mathbf{C}(\boldsymbol{\theta}) = \mathbf{I}.$$

Differentiating each element of the matrices and expressing in matrix form, we have

$$\mathbf{C}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k} + \frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_k} \mathbf{C}(\boldsymbol{\theta}) = \mathbf{0}$$

which leads to the desired result.

We are now ready to evaluate the CRLB. Taking the first derivative

$$\frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta_k} = -\frac{1}{2} \frac{\partial \ln \det[\mathbf{C}(\boldsymbol{\theta})]}{\partial \theta_k} - \frac{1}{2} \frac{\partial}{\partial \theta_k} \left[(\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta}))^T \mathbf{C}^{-1}(\boldsymbol{\theta}) (\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta})) \right].$$

The first term has already been evaluated using (3C.1), so that we consider the second term:

$$\frac{\partial}{\partial \theta_k} \left[(\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta}))^T \mathbf{C}^{-1}(\boldsymbol{\theta}) (\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta})) \right]$$

$$
\begin{aligned}
&= \frac{\partial}{\partial \theta_k} \sum_{i=1}^{N} \sum_{j=1}^{N} (x[i] - [\boldsymbol{\mu}(\boldsymbol{\theta})]_i)[\mathbf{C}^{-1}(\boldsymbol{\theta})]_{ij}(x[j] - [\boldsymbol{\mu}(\boldsymbol{\theta})]_j) \\
&= \sum_{i=1}^{N} \sum_{j=1}^{N} \left\{ (x[i] - [\boldsymbol{\mu}(\boldsymbol{\theta})]_i) \left[[\mathbf{C}^{-1}(\boldsymbol{\theta})]_{ij} \left(-\frac{\partial [\boldsymbol{\mu}(\boldsymbol{\theta})]_j}{\partial \theta_k} \right) \right. \right. \\
&\qquad \left. + \frac{\partial [\mathbf{C}^{-1}(\boldsymbol{\theta})]_{ij}}{\partial \theta_k} (x[j] - [\boldsymbol{\mu}(\boldsymbol{\theta})]_j) \right] \\
&\qquad \left. + \left(-\frac{\partial [\boldsymbol{\mu}(\boldsymbol{\theta})]_i}{\partial \theta_k} \right) [\mathbf{C}^{-1}(\boldsymbol{\theta})]_{ij}(x[j] - [\boldsymbol{\mu}(\boldsymbol{\theta})]_j) \right\} \\
&= -(\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta}))^T \mathbf{C}^{-1}(\boldsymbol{\theta}) \frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_k} + (\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta}))^T \frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_k} (\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta})) \\
&\quad - \frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})^T}{\partial \theta_k} \mathbf{C}^{-1}(\boldsymbol{\theta})(\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta})) \\
&= -2 \frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})^T}{\partial \theta_k} \mathbf{C}^{-1}(\boldsymbol{\theta})(\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta})) + (\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta}))^T \frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_k} (\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta})).
\end{aligned}
$$

Using (3C.1) and the last result, we have

$$
\begin{aligned}
\frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta_k} &= -\frac{1}{2} \mathrm{tr} \left(\mathbf{C}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k} \right) + \frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_k}^T \mathbf{C}^{-1}(\boldsymbol{\theta})(\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta})) \\
&\quad - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta}))^T \frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_k} (\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta})). \qquad \text{(3C.5)}
\end{aligned}
$$

Let $\mathbf{y} = \mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\theta})$. Evaluating

$$
[\mathbf{I}(\boldsymbol{\theta})]_{kl} = E \left[\frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta_k} \frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta_l} \right]
$$

which is equivalent to (3.23), yields

$$
\begin{aligned}
[\mathbf{I}(\boldsymbol{\theta})]_{kl} &= \frac{1}{4} \mathrm{tr} \left(\mathbf{C}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k} \right) \mathrm{tr} \left(\mathbf{C}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_l} \right) \\
&\quad + \frac{1}{2} \mathrm{tr} \left(\mathbf{C}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k} \right) E \left(\mathbf{y}^T \frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_l} \mathbf{y} \right) \\
&\quad + \frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_k}^T \mathbf{C}^{-1}(\boldsymbol{\theta}) E[\mathbf{y}\mathbf{y}^T] \mathbf{C}^{-1}(\boldsymbol{\theta}) \frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_l} \\
&\quad + \frac{1}{4} E \left[\mathbf{y}^T \frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_k} \mathbf{y}\mathbf{y}^T \frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_l} \mathbf{y} \right]
\end{aligned}
$$

where we note that all odd order moments are zero. Continuing, we have

$$[\mathbf{I}(\boldsymbol{\theta})]_{kl} =$$

$$\frac{1}{4}\mathrm{tr}\left(\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k}\right)\mathrm{tr}\left(\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_l}\right)$$

$$-\frac{1}{2}\mathrm{tr}\left(\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k}\right)\mathrm{tr}\left(\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_l}\right)$$

$$+\frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_k}^T \mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_l} + \frac{1}{4}E\left[\mathbf{y}^T\frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_k}\mathbf{y}\mathbf{y}^T\frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_l}\mathbf{y}\right] \tag{3C.6}$$

where $E(\mathbf{y}^T\mathbf{z}) = \mathrm{tr}[E(\mathbf{z}\mathbf{y}^T)]$ for $\mathbf{y},\mathbf{z}$ $N \times 1$ vectors and (3C.2) have been used. To evaluate the last term we use [Porat and Friedlander 1986]

$$E[\mathbf{y}^T\mathbf{A}\mathbf{y}\mathbf{y}^T\mathbf{B}\mathbf{y}] = \mathrm{tr}(\mathbf{A}\mathbf{C})\mathrm{tr}(\mathbf{B}\mathbf{C}) + 2\mathrm{tr}(\mathbf{A}\mathbf{C}\mathbf{B}\mathbf{C})$$

where $\mathbf{C} = E(\mathbf{y}\mathbf{y}^T)$ and $\mathbf{A}$ and $\mathbf{B}$ are symmetric matrices. Thus, this term becomes

$$\frac{1}{4}\mathrm{tr}\left(\frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_k}\mathbf{C}(\boldsymbol{\theta})\right)\mathrm{tr}\left(\frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_l}\mathbf{C}(\boldsymbol{\theta})\right)$$

$$+\frac{1}{2}\mathrm{tr}\left(\frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_k}\mathbf{C}(\boldsymbol{\theta})\frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_l}\mathbf{C}(\boldsymbol{\theta})\right).$$

Next, using the relationship (3C.2), this term becomes

$$\frac{1}{4}\mathrm{tr}\left(\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k}\right)\mathrm{tr}\left(\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_l}\right)$$

$$+\frac{1}{2}\mathrm{tr}\left(\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k}\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_l}\right) \tag{3C.7}$$

and finally, using (3C.7) in (3C.6), produces the desired result.

Appendix 3D

Derivation of Asymptotic CRLB

It can be proven that almost any WSS Gaussian random process $x[n]$ may be represented as the output of a causal linear shift invariant filter driven at the input by white Gaussian noise $u[n]$ [Brockwell and Davis 1987] or

$$x[n] = \sum_{k=0}^{\infty} h[k]u[n-k] \tag{3D.1}$$

where $h[0] = 1$. The only condition is that the PSD must satisfy

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \ln P_{xx}(f)\, df > -\infty.$$

With this representation the PSD of $x[n]$ is

$$P_{xx}(f) = |H(f)|^2 \sigma_u^2$$

where σ_u^2 is the variance of $u[n]$ and $H(f) = \sum_{k=0}^{\infty} h[k] \exp(-j2\pi fk)$ is the filter frequency response. If the observations are $\{x[0], x[1], \ldots, x[N-1]\}$ and N is large, then the representation is approximated by

$$\begin{aligned} x[n] &= \sum_{k=0}^{n} h[k]u[n-k] + \sum_{k=n+1}^{\infty} h[k]u[n-k] \\ &\approx \sum_{k=0}^{n} h[k]u[n-k]. \end{aligned} \tag{3D.2}$$

This is equivalent to setting $u[n] = 0$ for $n < 0$. As $n \to \infty$, the approximate representation becomes better for $x[n]$. It is clear, however, that the beginning samples will be poorly represented unless the impulse response $h[k]$ is small for $k > n$. For large N most of the samples will be accurately represented if N is much greater than the impulse response length. Since

$$r_{xx}[k] = \sigma_u^2 \sum_{n=0}^{\infty} h[n]h[n+k]$$

the *correlation time* or effective duration of $r_{xx}[k]$ is the same as the impulse response length. Hence, because the CRLB to be derived is based on (3D.2), *the asymptotic CRLB will be a good approximation if the data record length is much greater than the correlation time.*

To find the PDF of $\mathbf{x}$ we use (3D.2), which is a transformation from $\mathbf{u} = [u[0]\, u[1] \ldots u[N-1]]^T$ to $\mathbf{x} = [x[0]\, x[1] \ldots x[N-1]]^T$ or

$$\mathbf{x} = \underbrace{\begin{bmatrix} h[0] & 0 & 0 & \ldots & 0 \\ h[1] & h[0] & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ h[N-1] & h[N-2] & h[N-3] & \ldots & h[0] \end{bmatrix}}_{\mathbf{H}} \mathbf{u}.$$

Note that $\mathbf{H}$ has a determinant of $(h[0])^N = 1$ and hence is invertible. Since $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \sigma_u^2\mathbf{I})$, the PDF of $\mathbf{x}$ is $\mathcal{N}(\mathbf{0}, \sigma_u^2\mathbf{H}\mathbf{H}^T)$ or

$$p(\mathbf{x};\boldsymbol{\theta}) = \frac{1}{(2\pi)^{\frac{N}{2}} \det^{\frac{1}{2}}(\sigma_u^2\mathbf{H}\mathbf{H}^T)} \exp\left[-\frac{1}{2}\mathbf{x}^T(\sigma_u^2\mathbf{H}\mathbf{H}^T)^{-1}\mathbf{x}\right].$$

But

$$\det(\sigma_u^2\mathbf{H}\mathbf{H}^T) = \sigma_u^{2N}\det^2(\mathbf{H}) = \sigma_u^{2N}.$$

Also,

$$\mathbf{x}^T(\sigma_u^2\mathbf{H}\mathbf{H}^T)^{-1}\mathbf{x} = \frac{1}{\sigma_u^2}(\mathbf{H}^{-1}\mathbf{x})^T(\mathbf{H}^{-1}\mathbf{x}) = \frac{1}{\sigma_u^2}\mathbf{u}^T\mathbf{u}$$

so that

$$p(\mathbf{x};\boldsymbol{\theta}) = \frac{1}{(2\pi\sigma_u^2)^{\frac{N}{2}}} \exp\left(-\frac{1}{2\sigma_u^2}\mathbf{u}^T\mathbf{u}\right). \tag{3D.3}$$

From (3D.2) we have approximately

$$X(f) = H(f)U(f)$$

where

$$\begin{aligned} X(f) &= \sum_{n=0}^{N-1} x[n]\exp(-j2\pi fn) \\ U(f) &= \sum_{n=0}^{N-1} u[n]\exp(-j2\pi fn) \end{aligned}$$

are the Fourier transforms of the truncated sequences. By Parseval's theorem

$$\frac{1}{\sigma_u^2}\mathbf{u}^T\mathbf{u} = \frac{1}{\sigma_u^2}\sum_{n=0}^{N-1} u^2[n]$$

$$\begin{aligned} &= \frac{1}{\sigma_u^2}\int_{-\frac{1}{2}}^{\frac{1}{2}} |U(f)|^2 df \\ &\approx \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{|X(f)|^2}{\sigma_u^2 |H(f)|^2}\, df \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{|X(f)|^2}{P_{xx}(f)}\, df. \end{aligned} \tag{3D.4}$$

Also,

$$\begin{aligned} \ln \sigma_u^2 &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \ln \sigma_u^2\, df \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \ln\left(\frac{P_{xx}(f)}{|H(f)|^2}\right) df \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \ln P_{xx}(f)\, df - \int_{-\frac{1}{2}}^{\frac{1}{2}} \ln |H(f)|^2 df. \end{aligned}$$

But

$$\begin{aligned} \int_{-\frac{1}{2}}^{\frac{1}{2}} \ln |H(f)|^2 df &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \ln H(f) + \ln H^*(f)\, df \\ &= 2\,\mathrm{Re}\int_{-\frac{1}{2}}^{\frac{1}{2}} \ln H(f)\, df \\ &= 2\,\mathrm{Re}\oint_C \ln \mathcal{H}(z)\frac{dz}{2\pi j z} \\ &= 2\,\mathrm{Re}\left[\mathcal{Z}^{-1}\left\{\ln \mathcal{H}(z)\right\}\big|_{n=0}\right] \end{aligned}$$

where C is the unit circle in the z plane. Since $\mathcal{H}(z)$ corresponds to the system function of a causal filter, it converges outside a circle of radius $r < 1$ (since $\mathcal{H}(z)$ is assumed to exist on the unit circle for the frequency response to exist). Hence, $\ln \mathcal{H}(z)$ also converges outside a circle of radius $r < 1$, so that the corresponding sequence is causal. By the initial value theorem which is valid for a causal sequence

$$\begin{aligned} \mathcal{Z}^{-1}\{\ln \mathcal{H}(z)\}|_{n=0} &= \lim_{z\to\infty} \ln \mathcal{H}(z) \\ &= \ln \lim_{z\to\infty} \mathcal{H}(z) \\ &= \ln h[0] = 0. \end{aligned}$$

Therefore,

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \ln |H(f)|^2 df = 0$$

and finally

$$\ln \sigma_u^2 = \int_{-\frac{1}{2}}^{\frac{1}{2}} \ln P_{xx}(f)\, df. \tag{3D.5}$$

Substituting (3D.4) and (3D.5) into (3D.3) produces for the log PDF

$$\ln p(\mathbf{x};\boldsymbol{\theta}) = -\frac{N}{2}\ln 2\pi - \frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \ln P_{xx}(f)\, df - \frac{1}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{|X(f)|^2}{P_{xx}(f)}\, df.$$

Hence, the asymptotic log PDF is

$$\ln p(\mathbf{x};\boldsymbol{\theta}) = -\frac{N}{2}\ln 2\pi - \frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \left[\ln P_{xx}(f) + \frac{\frac{1}{N}|X(f)|^2}{P_{xx}(f)}\right] df. \tag{3D.6}$$

To determine the CRLB

$$\begin{aligned}
\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_i} &= -\frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{1}{P_{xx}(f)} - \frac{\frac{1}{N}|X(f)|^2}{P_{xx}^2(f)}\right) \frac{\partial P_{xx}(f)}{\partial \theta_i}\, df \\
\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j} &= -\frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{1}{P_{xx}(f)} - \frac{\frac{1}{N}|X(f)|^2}{P_{xx}^2(f)}\right) \frac{\partial^2 P_{xx}(f)}{\partial \theta_i \partial \theta_j} \\
&\quad + \left(-\frac{1}{P_{xx}^2(f)} + \frac{\frac{2}{N}|X(f)|^2}{P_{xx}^3(f)}\right) \frac{\partial P_{xx}(f)}{\partial \theta_i}\frac{\partial P_{xx}(f)}{\partial \theta_j}\, df.
\end{aligned} \tag{3D.7}$$

In taking the expected value we encounter the term $E(|X(f)|^2/N)$. For large N this is now shown to be $P_{xx}(f)$. Note that $|X(f)|^2/N$ is termed the *periodogram* spectral estimator.

$$\begin{aligned}
E\left(\frac{1}{N}|X(f)|^2\right) &= E\left(\frac{1}{N}\sum_{m=0}^{N-1}\sum_{n=0}^{N-1} x[m]x[n]\exp[-j2\pi f(m-n)]\right) \\
&= \frac{1}{N}\sum_{m=0}^{N-1}\sum_{n=0}^{N-1} r_{xx}[m-n]\exp[-j2\pi f(m-n)] \\
&= \sum_{k=-(N-1)}^{N-1}\left(1-\frac{|k|}{N}\right) r_{xx}[k]\exp(-j2\pi f k)
\end{aligned} \tag{3D.8}$$

where we have used the identity

$$\sum_{m=0}^{N-1}\sum_{n=0}^{N-1} g[m-n] = \sum_{k=-(N-1)}^{N-1} (N-|k|)g[k].$$

As $N \to \infty$,

$$\left(1 - \frac{|k|}{N}\right) r_{xx}[k] \to r_{xx}[k]$$

assuming that the ACF dies out sufficiently rapidly. Hence,

$$E\left[\frac{1}{N}|X(f)|^2\right] \approx P_{xx}(f).$$

Upon taking expectations in (3D.7), the first term is zero, and finally,

$$\begin{aligned}
[\mathbf{I}(\boldsymbol{\theta})]_{ij} &= \frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{P_{xx}^2(f)} \frac{\partial P_{xx}(f)}{\partial \theta_i} \frac{\partial P_{xx}(f)}{\partial \theta_j}\, df \\
&= \frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{\partial \ln P_{xx}(f)}{\partial \theta_i} \frac{\partial \ln P_{xx}(f)}{\partial \theta_j}\, df
\end{aligned}$$

which is (3.34) without the explicit dependence of the PSD on $\boldsymbol{\theta}$ shown.